

Physics Laws & Equations

Master Compendium

V4 - rebuilt as a proper long-form PDF, targeting 80-100 pages.

Includes classical mechanics, Newton laws, gravitation, thermodynamics, gas laws, electromagnetism, circuits, optics, waves, fluids, relativity, quantum mechanics, atomic/nuclear/particle physics, plasma, materials, astrophysics, physical chemistry and mathematics used in physics.

Important: there is no official final count of every physics law. This document treats laws, principles, equations, theorems, postulates and named rules as the practical universe of physics law references.

Prepared for Sam Kazangas

Dense equation/law reference edition - not a 14-page summary.

Contents / Scope Map

1. Scope, counting rules and universal structure (8 entries)
2. Classical mechanics and Newtonian laws (49 entries)
3. Gravitation, celestial mechanics and orbital laws (24 entries)
4. Thermodynamics, heat, entropy and transport laws (47 entries)
5. Gas laws and kinetic theory (21 entries)
6. Electromagnetism, Maxwell laws and field equations (44 entries)
7. Electrical circuits, electronics and information laws (35 entries)
8. Waves, sound, optics and radiation laws (41 entries)
9. Fluid mechanics, aerodynamics and hydrodynamic laws (35 entries)
10. Relativity, spacetime and cosmology equations (29 entries)
11. Quantum mechanics, atomic laws and quantum information (55 entries)
12. Nuclear physics, particle physics and quantum field laws (25 entries)
13. Solid state, materials science and condensed matter laws (30 entries)
14. Plasma physics and magnetohydrodynamic laws (20 entries)
15. Astrophysics and cosmology laws (20 entries)
16. Physical chemistry laws used in physics (24 entries)
17. Mathematical laws and theorems used in physics (31 entries)
18. Mechanics formula appendix - kinematics, rotation and rigid bodies (18 entries)
19. Electromagnetic and circuit formula appendix (25 entries)
20. Quantum equation appendix - operators, spectra and information (20 entries)
21. Relativity and cosmology formula appendix (18 entries)
22. Statistical mechanics, probability and data laws (20 entries)
23. Constants, units and dimensional reference (20 entries)
24. Alphabetic named law checklist - additional physics/engineering laws (46 entries)
25. Extended equation bank - harmonics, geometry, scaling and dynamics (96 entries)

Scope, counting rules and universal structure

No single official count

Eq: strict laws ~ 30-60; major laws/equations ~ 400-700; specialised correlations > 1000

Law/use: Physics has no world registry of laws; this compendium includes laws, principles, postulates, theorems and named equations used as laws.

Dimensions of mechanics

Eq: [M], [L], [T]

Law/use: Mass, length and time form the base dimensions for most classical mechanics.

Dimensions of charge

Eq: [Q] = [I*T]

Law/use: Electric charge is current integrated over time.

Dimensional homogeneity

Eq: each term in an equation must have the same dimensions

Law/use: A core consistency law behind all physical formula checking.

SI base units

Eq: m, kg, s, A, K, mol, cd

Law/use: Metre, kilogram, second, ampere, kelvin, mole and candela.

Superposition principle

Eq: response(total) = sum response(parts)

Law/use: Linear systems allow independent contributions to add; used in waves, fields, circuits and quantum states.

Locality principle

Eq: local interactions propagate no faster than c

Law/use: Modern relativistic field theory is built around local causal structure.

Symmetry principle

Eq: symmetry -> conservation law

Law/use: Noether links invariances to conserved quantities.

Classical mechanics and Newtonian laws

Newton first law

Eq: $\sum F = 0 \rightarrow v = \text{constant}$

Law/use: A body remains at rest or constant velocity unless acted on by net external force.

Newton second law

Eq: $\sum F = m \cdot a = d p / dt$

Law/use: Net force equals rate of change of momentum; for constant mass $F = ma$.

Newton third law

Eq: $F_{AB} = -F_{BA}$

Law/use: Forces between two interacting bodies are equal in magnitude and opposite in direction.

Newton gravitation

Eq: $F = G m_1 m_2 / r^2$

Law/use: Every mass attracts every other mass with inverse-square force.

Inertia

Eq: $p = m \cdot v$

Law/use: Inertial mass measures resistance to acceleration.

Work

Eq: $W = \int F \cdot ds$

Law/use: Work transfers energy through force over displacement.

Power

Eq: $P = dW/dt = F \cdot v$

Law/use: Power is the rate of work or energy transfer.

Kinetic energy

Eq: $K = 1/2 \cdot m \cdot v^2$

Law/use: Energy associated with translational motion.

Potential energy near Earth

Eq: $U = m \cdot g \cdot h$

Law/use: Gravitational potential energy for small height changes.

Spring potential energy

Eq: $U = 1/2 \cdot k \cdot x^2$

Law/use: Energy stored in a linear spring.

Hooke law

Eq: $F = -k \cdot x$

Law/use: Linear restoring force proportional to displacement.

Impulse momentum theorem

Eq: $J = \int F dt = \Delta p$

Law/use: Impulse equals change in momentum.

Work energy theorem

Eq: $W_{\text{net}} = \Delta K$

Law/use: Net work changes kinetic energy.

Torque

Eq: $\tau = r \times F$

Law/use: Rotational effect of a force about an axis.

Angular momentum

Eq: $L = r \times p$

Law/use: Rotational analogue of linear momentum.

Rotational Newton law

$$\text{Eq: } \sum \tau = I \alpha$$

Law/use: Net torque equals moment of inertia times angular acceleration.

Moment of inertia

$$\text{Eq: } I = \sum m_i r_i^2 = \int r^2 dm$$

Law/use: Mass distribution resistance to angular acceleration.

Rotational kinetic energy

$$\text{Eq: } K_{\text{rot}} = \frac{1}{2} I \omega^2$$

Law/use: Energy of rigid-body rotation.

Rolling without slipping

$$\text{Eq: } v_{\text{cm}} = \omega R$$

Law/use: Condition for pure rolling contact.

Centripetal acceleration

$$\text{Eq: } a_c = v^2/r = \omega^2 r$$

Law/use: Acceleration directed toward centre of circular path.

Centripetal force

$$\text{Eq: } F_c = m v^2/r$$

Law/use: Required inward force for circular motion.

Friction static

$$\text{Eq: } f_s \leq \mu_s N$$

Law/use: Static friction adjusts up to a maximum.

Friction kinetic

$$\text{Eq: } f_k = \mu_k N$$

Law/use: Sliding friction model proportional to normal force.

Drag low Reynolds

$$\text{Eq: } F_d = 6 \pi \eta r v$$

Law/use: Stokes drag for small spheres in viscous flow.

Drag high speed

$$\text{Eq: } F_d = \frac{1}{2} \rho C_d A v^2$$

Law/use: Quadratic aerodynamic drag.

Terminal velocity

$$\text{Eq: } mg = \text{drag} + \text{buoyancy}$$

Law/use: Maximum falling speed when forces balance.

D Alembert principle

$$\text{Eq: } \sum (F_i - m_i a_i) \delta r_i = 0$$

Law/use: Dynamics converted to virtual-work equilibrium form.

Virtual work principle

$$\text{Eq: } \delta W = 0$$

Law/use: Equilibrium condition under virtual displacements.

Lagrange equations

$$\text{Eq: } \frac{d}{dt} \left(\frac{dL}{dq_i} \right) - \frac{dL}{dq_i} = Q_i$$

Law/use: Generalised equations of motion.

Euler-Lagrange equation

$$\text{Eq: } \frac{d}{dt} \left(\frac{dL}{dq} \right) - \frac{dL}{dq} = 0$$

Law/use: Stationary action produces dynamics.

Hamilton equations

$$\text{Eq: } \dot{q}_i = \frac{dH}{dp_i}; \quad \dot{p}_i = -\frac{dH}{dq_i}$$

Law/use: Canonical phase-space dynamics.

Hamilton principle

$$\text{Eq: } \delta S = 0; \quad S = \int L dt$$

Law/use: Actual path extremises the action.

Poisson bracket evolution

$$\text{Eq: } df/dt = \{f, H\} + df/dt_{\text{explicit}}$$

Law/use: Classical observable evolution in Hamiltonian mechanics.

Liouville theorem

$$\text{Eq: } d \rho_{\text{phase}}/dt = 0$$

Law/use: Phase-space volume is conserved under Hamiltonian flow.

Noether theorem

$$\text{Eq: continuous symmetry} \rightarrow \text{conserved current}$$

Law/use: Time symmetry gives energy; space symmetry gives momentum; rotation symmetry gives angular momentum.

Virial theorem

$$\text{Eq: } 2\langle K \rangle = \langle r \cdot dU/dr \rangle$$

Law/use: Relates average kinetic and potential energy in bound systems.

Parallel axis theorem

$$\text{Eq: } I = I_{\text{cm}} + M d^2$$

Law/use: Moment of inertia about shifted parallel axis.

Perpendicular axis theorem

$$\text{Eq: } I_z = I_x + I_y$$

Law/use: For planar lamina about perpendicular axis.

Coefficient of restitution

$$\text{Eq: } e = \text{relative speed after} / \text{relative speed before}$$

Law/use: Measures elasticity of impact.

Elastic collision 1D

Eq: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$; $K_{\text{before}} = K_{\text{after}}$

Law/use: Momentum and kinetic energy conserved.

Inelastic collision

Eq: momentum conserved; kinetic energy not conserved

Law/use: Part of kinetic energy converts to heat/deformation.

Center of mass

Eq: $R_{\text{cm}} = \sum m_i r_i / \sum m_i$

Law/use: Mass-weighted average position.

Center of mass motion

Eq: $M \ddot{r}_{\text{cm}} = F_{\text{ext}}$

Law/use: COM responds only to external net force.

Reduced mass

Eq: $\mu = m_1 m_2 / (m_1 + m_2)$

Law/use: Two-body problem reduces to one body of reduced mass.

Simple harmonic oscillator

Eq: $x(t) = A \cos(\omega t + \phi)$; $\omega = \sqrt{k/m}$

Law/use: Linear oscillator law.

Pendulum small angle

Eq: $T = 2\pi \sqrt{L/g}$

Law/use: Period of simple pendulum for small angles.

Torsion oscillator

Eq: $T = 2\pi \sqrt{I/\kappa}$

Law/use: Rotational oscillation period.

Projectile range level ground

Eq: $R = v_0^2 \sin(2\theta)/g$

Law/use: Range without drag for equal launch and landing height.

Escape velocity

Eq: $v_{\text{esc}} = \sqrt{2GM/r}$

Law/use: Minimum speed to escape gravitational field neglecting drag.

Gravitation, celestial mechanics and orbital laws**Kepler first law**

Eq: orbit is ellipse with central body at one focus

Law/use: Planetary bound orbit geometry.

Kepler second law

Eq: $dA/dt = \text{constant}$

Law/use: Equal areas swept in equal times; angular momentum conservation.

Kepler third law

Eq: $T^2 = 4\pi^2 a^3 / (GM)$

Law/use: Period-squared proportional to semi-major-axis cubed.

Orbital speed circular

Eq: $v = \sqrt{GM/r}$

Law/use: Circular orbit speed.

Orbital energy

Eq: $E = -GMm/(2a)$

Law/use: Specific bound orbit energy depends on semi-major axis.

Vis-viva equation

Eq: $v^2 = GM(2/r - 1/a)$

Law/use: Speed anywhere on a Kepler orbit.

Gravitational potential

Eq: $\Phi = -GM/r$

Law/use: Potential per unit mass outside spherical mass.

Gravitational field

Eq: $g = -\text{grad } \Phi$

Law/use: Acceleration field from potential.

Gauss law for gravity

Eq: $\oint g \cdot dA = -4\pi G M_{\text{enclosed}}$

Law/use: Flux of gravity field through closed surface.

Poisson gravity equation

Eq: $\nabla^2 \Phi = 4\pi G \rho$

Law/use: Potential sourced by mass density.

Laplace gravity equation

Eq: $\nabla^2 \Phi = 0$

Law/use: Potential in source-free region.

Shell theorem

Eq: spherical shell acts as point mass outside; zero field inside shell

Law/use: Newtonian result for spherical symmetry.

Roche limit fluid

Eq: $d \sim 2.44 R (\rho_M / \rho_m)^{1/3}$

Law/use: Tidal disruption distance estimate.

Hill radius

Eq: $r_H = a (m / (3M))^{1/3}$

Law/use: Region where smaller body dominates satellites.

Jeans length

$$\text{Eq: } \lambda_{\text{J}} = c_s \sqrt{\pi / (G \rho)}$$

Law/use: Scale above which gravity overcomes pressure support.

Jeans mass

$$\text{Eq: } M_{\text{J}} \sim \rho \lambda_{\text{J}}^3$$

Law/use: Mass threshold for gravitational collapse.

Chandrasekhar limit

$$\text{Eq: } M_{\text{Ch}} \sim 1.4 \text{ solar masses}$$

Law/use: Maximum white dwarf mass supported by electron degeneracy.

Eddington luminosity

$$\text{Eq: } L_{\text{Edd}} = 4 \pi G M m_p c / \sigma_T$$

Law/use: Radiation pressure balances gravity for ionized gas.

Virial for gravity

$$\text{Eq: } 2K + U = 0$$

Law/use: Bound gravitational system equilibrium average.

Tidal acceleration

$$\text{Eq: } a_{\text{tidal}} \sim 2GM/R/r^3$$

Law/use: Differential gravitational pull across size R.

Lagrange points

$$\text{Eq: } \text{grad effective potential} = 0$$

Law/use: Equilibrium points in restricted three-body problem.

Binet equation

$$\text{Eq: } d^2u/d\theta^2 + u = -F(1/u)/(m h^2 u^2)$$

Law/use: Central-force orbit equation.

Precession condition

$$\text{Eq: } \text{closed orbits for } 1/r \text{ and } r^2 \text{ potentials}$$

Law/use: Bertrand theorem result.

Newtonian redshift approximation

$$\text{Eq: } \Delta f/f \sim \Delta \Phi/c^2$$

Law/use: Weak-field gravitational frequency shift.

Thermodynamics, heat, entropy and transport laws**Zeroth law thermodynamics**

$$\text{Eq: } \text{if A in thermal equilibrium with B and B with C, then A with C}$$

Law/use: Defines temperature as a state variable.

First law thermodynamics

$$\text{Eq: } \Delta U = Q - W$$

Law/use: Energy conservation for thermodynamic systems.

Second law thermodynamics

$$\text{Eq: } \Delta S_{\text{universe}} \geq 0$$

Law/use: Entropy of isolated system never decreases.

Third law thermodynamics

$$\text{Eq: } S \rightarrow \text{constant as } T \rightarrow 0 \text{ K}$$

Law/use: Perfect crystal entropy tends to zero in ideal statement.

Clausius inequality

$$\text{Eq: } \oint \delta Q/T \leq 0$$

Law/use: Cycle inequality defining entropy.

Carnot efficiency

$$\text{Eq: } \eta_{\text{max}} = 1 - T_c/T_h$$

Law/use: Maximum heat-engine efficiency between reservoirs.

Heat pump COP

$$\text{Eq: } \text{COP}_{\text{heating}} = Q_h/W$$

Law/use: Coefficient of performance for heating.

Refrigerator COP

$$\text{Eq: } \text{COP}_{\text{cooling}} = Q_c/W$$

Law/use: Coefficient of performance for cooling.

Entropy definition reversible

$$\text{Eq: } dS = \delta Q_{\text{rev}}/T$$

Law/use: Entropy change for reversible heat transfer.

Boltzmann entropy

$$\text{Eq: } S = k_B \ln \Omega$$

Law/use: Entropy from number of microstates.

Gibbs entropy

$$\text{Eq: } S = -k_B \sum p_i \ln p_i$$

Law/use: Statistical entropy for probabilities.

Helmholtz free energy

$$\text{Eq: } F = U - TS$$

Law/use: Useful work potential at constant temperature and volume.

Gibbs free energy

$$\text{Eq: } G = H - TS$$

Law/use: Spontaneity criterion at constant temperature and pressure.

Enthalpy

$$\text{Eq: } H = U + pV$$

Law/use: Thermodynamic potential useful at constant pressure.

Thermodynamic identity

$$\text{Eq: } dU = T dS - p dV + \mu dN$$

Law/use: Core differential relation.

Maxwell relation 1

$$\text{Eq: } (dT/dV)_S = -(dp/dS)_V$$

Law/use: From equality of mixed partials.

Maxwell relation 2

$$\text{Eq: } (dT/dp)_S = (dV/dS)_p$$

Law/use: From enthalpy.

Maxwell relation 3

$$\text{Eq: } (dS/dV)_T = (dp/dT)_V$$

Law/use: From Helmholtz energy.

Maxwell relation 4

$$\text{Eq: } (dS/dp)_T = -(dV/dT)_p$$

Law/use: From Gibbs energy.

Gibbs-Duhem equation

$$\text{Eq: } S dT - V dp + \sum N_i d\mu_i = 0$$

Law/use: Intensive variables are not independent.

Gibbs phase rule

$$\text{Eq: } F = C - P + 2$$

Law/use: Degrees of freedom for phases and components.

Clapeyron equation

$$\text{Eq: } dp/dT = \Delta S / \Delta V$$

Law/use: Slope of phase boundary.

Clausius-Clapeyron

$$\text{Eq: } d \ln p / dT = L / (R T^2)$$

Law/use: Vapor-pressure temperature dependence approximation.

Heat capacity constant volume

$$\text{Eq: } C_V = (dU/dT)_V$$

Law/use: Energy needed to raise temperature at constant volume.

Heat capacity constant pressure

$$\text{Eq: } C_p = (dH/dT)_p$$

Law/use: Heat needed to raise temperature at constant pressure.

Mayer relation ideal gas

$$\text{Eq: } C_p - C_V = R$$

Law/use: Ideal-gas heat capacity difference per mole.

Adiabatic ideal gas

$$\text{Eq: } pV^\gamma = \text{constant}$$

Law/use: Reversible adiabatic ideal gas law.

Adiabatic temperature law

$$\text{Eq: } T V^{(\gamma-1)} = \text{constant}$$

Law/use: Temperature-volume law for ideal gas adiabatic.

Adiabatic pressure-temp law

$$\text{Eq: } T^\gamma p^{(1-\gamma)} = \text{constant}$$

Law/use: Pressure-temperature adiabatic relation.

Equipartition theorem

$$\text{Eq: } \langle E \rangle = (f/2) k_B T$$

Law/use: Each quadratic degree contributes half kT.

Dulong-Petit law

$$\text{Eq: } C_V \sim 3R$$

Law/use: High-temperature molar heat capacity of solids.

Debye T cubed law

$$\text{Eq: } C_V \text{ proportional } T^3$$

Law/use: Low-temperature lattice heat capacity.

Fourier heat conduction

$$\text{Eq: } q = -k \text{ grad } T$$

Law/use: Heat flux flows down temperature gradient.

Thermal diffusion equation

$$\text{Eq: } dT/dt = \alpha \nabla^2 T$$

Law/use: Heat equation for constant properties.

Newton cooling law

$$\text{Eq: } dT/dt = -k(T - T_{\text{env}})$$

Law/use: Convective cooling proportional to temperature difference.

Stefan-Boltzmann law

$$\text{Eq: } j_{\text{star}} = \sigma T^4$$

Law/use: Blackbody radiated power per area.

Wien displacement law

$$\text{Eq: } \lambda_{\text{max}} T = b$$

Law/use: Peak blackbody wavelength relation.

Planck radiation law

$$\text{Eq: } B_{\nu} = 2h\nu^3/c^2 / (\exp(h\nu/kT) - 1)$$

Law/use: Spectral radiance of blackbody.

Rayleigh-Jeans law

Eq: $B_{\nu} = 2\nu^2 kT/c^2$

Law/use: Low-frequency blackbody approximation.

Kirchhoff thermal radiation law

Eq: emissivity = absorptivity at thermal equilibrium

Law/use: Good absorbers are good emitters.

Nernst heat theorem

Eq: $\Delta S \rightarrow 0$ as $T \rightarrow 0$ for reactions

Law/use: A form of the third law.

Joule-Thomson coefficient

Eq: $\mu_{JT} = (dT/dp)_H$

Law/use: Temperature change under throttling.

Onsager reciprocal relations

Eq: $L_{ij} = L_{ji}$

Law/use: Near-equilibrium reciprocal transport coefficients.

Fluctuation-dissipation theorem

Eq: response relates to equilibrium fluctuations

Law/use: Dissipation and noise are linked.

Landauer principle

Eq: $E_{\min} = k_B T \ln 2$ per erased bit

Law/use: Information erasure has thermodynamic cost.

Jarzynski equality

Eq: $\langle \exp(-\beta W) \rangle = \exp(-\beta \Delta F)$

Law/use: Non-equilibrium work relation.

Crooks fluctuation theorem

Eq: $P_F(W)/P_R(-W) = \exp(\beta(W - \Delta F))$

Law/use: Forward/reverse work distribution relation.

Gas laws and kinetic theory**Boyle law**

Eq: $pV = \text{constant}$ at fixed T, n

Law/use: Pressure inversely proportional to volume.

Charles law

Eq: $V/T = \text{constant}$ at fixed p, n

Law/use: Volume proportional to absolute temperature.

Gay-Lussac law

Eq: $p/T = \text{constant}$ at fixed V, n

Law/use: Pressure proportional to absolute temperature.

Avogadro law

Eq: $V/n = \text{constant}$ at fixed p, T

Law/use: Volume proportional to amount of gas.

Ideal gas law

Eq: $pV = nRT = N k_B T$

Law/use: Equation of state for ideal gas.

Combined gas law

Eq: $p_1 V_1/T_1 = p_2 V_2/T_2$

Law/use: Fixed amount of ideal gas.

Dalton partial pressures

Eq: $p_{\text{total}} = \sum p_i$

Law/use: Ideal gas mixture pressure sum.

Amagat law

Eq: $V_{\text{total}} = \sum V_i$

Law/use: Ideal gas mixture volume sum at same p, T .**Graham effusion law**

Eq: $\text{rate}_1/\text{rate}_2 = \sqrt{M_2/M_1}$

Law/use: Effusion rate inversely proportional to square root molar mass.

Henry law

Eq: $c = k_H p$

Law/use: Gas solubility proportional to partial pressure.

Raoult law

Eq: $p_i = x_i p_{i,\text{star}}$

Law/use: Ideal solution vapor pressure contribution.

Van der Waals equation

Eq: $(p + a n^2/V^2)(V - nb) = nRT$

Law/use: Real-gas correction for attraction and finite size.

Virial equation

Eq: $pV_m/(RT) = 1 + B/V_m + C/V_m^2 + \dots$

Law/use: Real gas expansion.

Maxwell-Boltzmann distribution

Eq: $f(v) = 4\pi (m/2\pi kT)^{3/2} v^2 \exp(-mv^2/2kT)$

Law/use: Gas molecular speed distribution.

Mean speed

Eq: $\langle v \rangle = \sqrt{8kT/(\pi m)}$

Law/use: Average molecular speed.

Most probable speed

Eq: $v_{mp} = \sqrt{2kT/m}$

Law/use: Peak of speed distribution.

RMS speed

Eq: $v_{rms} = \sqrt{3kT/m}$

Law/use: Root-mean-square speed.

Mean free path

Eq: $\lambda = 1/(\sqrt{2} \cdot n \cdot \pi \cdot d^2)$

Law/use: Average distance between collisions.

Pressure kinetic relation

Eq: $p = \frac{1}{3} \rho \langle v^2 \rangle$

Law/use: Gas pressure from particle momentum flux.

Boltzmann factor

Eq: $p_i \propto \exp(-E_i/kT)$

Law/use: Relative probability of energy state.

Saha equation

Eq: $n_{ion} n_e / n_{neutral} = \text{function}(T) \cdot \exp(-\chi/kT)$

Law/use: Ionization balance in gas/plasma.

Electromagnetism, Maxwell laws and field equations**Coulomb law**

Eq: $F = (1/(4\pi\epsilon_0)) \cdot q_1 q_2 / r^2$

Law/use: Electrostatic inverse-square force between charges.

Electric field definition

Eq: $E = F/q$

Law/use: Force per unit positive test charge.

Electric potential

Eq: $V = U/q$; $E = -\text{grad } V$

Law/use: Potential energy per charge.

Gauss law electricity

Eq: $\oint E \cdot dA = Q_{enclosed} / \epsilon_0$

Law/use: Electric flux equals enclosed charge over permittivity.

Gauss law magnetism

Eq: $\oint B \cdot dA = 0$

Law/use: No isolated magnetic monopoles in classical electromagnetism.

Faraday induction law

Eq: $\oint E \cdot dl = -d \Phi_B / dt$

Law/use: Changing magnetic flux creates circulating electric field.

Ampere-Maxwell law

Eq: $\oint B \cdot dl = \mu_0 I + \mu_0 \epsilon_0 d \Phi_E / dt$

Law/use: Currents and changing electric flux create magnetic circulation.

Maxwell equations differential

Eq: $\text{div } E = \rho / \epsilon_0$; $\text{div } B = 0$; $\text{curl } E = -dB/dt$; $\text{curl } B = \mu_0 J + \mu_0 \epsilon_0 dE/dt$

Law/use: Local field equations of electromagnetism.

Lorentz force

Eq: $F = q(E + v \times B)$

Law/use: Force on charge moving in electromagnetic fields.

Electromagnetic wave speed

Eq: $c = 1/\sqrt{\mu_0 \epsilon_0}$

Law/use: EM waves propagate at speed of light in vacuum.

Poynting vector

Eq: $S = (1/\mu_0) E \times B$

Law/use: Energy flux density of electromagnetic field.

EM energy density

Eq: $u = 1/2(\epsilon_0 E^2 + B^2/\mu_0)$

Law/use: Energy stored in electromagnetic field.

Continuity charge equation

Eq: $d\rho/dt + \text{div } J = 0$

Law/use: Local conservation of electric charge.

Ohm law

Eq: $V = I R$; $J = \sigma E$

Law/use: Linear relation between voltage/current and field/current density.

Resistivity

Eq: $R = \rho L/A$

Law/use: Resistance from material and geometry.

Joule heating

Eq: $P = I^2 R = V I = V^2/R$

Law/use: Electrical power dissipated as heat.

Kirchhoff current law

Eq: $\sum \text{currents into node} = \sum \text{currents out}$

Law/use: Charge conservation in circuits.

Kirchhoff voltage law

Eq: $\sum \text{voltage rises/drops around loop} = 0$

Law/use: Energy conservation in lumped circuits.

Capacitance

Eq: $C = Q/V$

Law/use: Charge stored per voltage.

Parallel plate capacitance

Eq: $C = \epsilon A/d$

Law/use: Capacitance of plate geometry.

Capacitor energy

Eq: $U = 1/2 C V^2 = Q^2/(2C)$

Law/use: Energy stored in electric field.

Inductance

Eq: $V_L = L dI/dt$

Law/use: Back-emf from changing current.

Inductor energy

Eq: $U = 1/2 L I^2$

Law/use: Energy stored in magnetic field.

Biot-Savart law

Eq: $dB = \mu_0/(4\pi) I dl \times \hat{r}/r^2$

Law/use: Magnetic field contribution from current element.

Ampere force wires

Eq: $F/L = \mu_0 I_1 I_2/(2\pi r)$

Law/use: Force per length between parallel currents.

Lenz law

Eq: induced current opposes change causing it

Law/use: Direction law for induction.

Faraday transformer emf

Eq: $\text{emf} = -N d\Phi_B/dt$

Law/use: Induced emf in coil.

Magnetic flux

Eq: $\Phi_B = \int B \cdot dA$

Law/use: Magnetic field through a surface.

Hall effect

Eq: $V_H = I B/(n q t)$

Law/use: Transverse voltage from current in magnetic field.

Cyclotron radius

Eq: $r = m v/(q B)$

Law/use: Radius of charged particle in uniform magnetic field.

Cyclotron frequency

Eq: $\omega_c = qB/m$

Law/use: Angular frequency in magnetic field.

Larmor radius

Eq: $r_L = m v_{\perp}/(qB)$

Law/use: Gyroradius of charged particle.

Larmor power

Eq: $P = q^2 a^2/(6\pi\epsilon_0 c^3)$

Law/use: Radiated power from accelerating nonrelativistic charge.

Aharonov-Bohm phase

Eq: $\Delta\phi = q \Phi_B/\hbar$

Law/use: Quantum phase from vector potential despite zero local B.

London equation

Eq: $\nabla^2 B = B/\lambda_L^2$

Law/use: Magnetic field expulsion in superconductor.

Josephson current

Eq: $I = I_c \sin \phi$

Law/use: Supercurrent through Josephson junction.

AC Josephson relation

Eq: $d\phi/dt = 2eV/\hbar$

Law/use: Voltage-phase evolution in junction.

Meissner effect

Eq: B expelled from superconductor

Law/use: Superconductors actively expel magnetic fields.

Curie law

Eq: $\chi = C/T$

Law/use: Paramagnetic susceptibility law.

Curie-Weiss law

Eq: $\chi = C/(T - T_C)$

Law/use: Magnetic susceptibility near ordering.

Gauss capacitance sphere

Eq: $C = 4\pi\epsilon_0 R$

Law/use: Isolated conducting sphere capacitance.

Skin depth

Eq: $\delta = \sqrt{2/(\omega \mu \sigma)}$

Law/use: AC penetration depth in conductor.

Poynting theorem

Eq: $\frac{d}{dt} \int_V u \, dV + \oint_V \mathbf{S} \cdot d\mathbf{A} = - \int_V \mathbf{J} \cdot \mathbf{E} \, dV$

Law/use: EM energy conservation law.

Maxwell stress tensor

Eq: force from field momentum flux

Law/use: Stress representation of electromagnetic forces.

Electrical circuits, electronics and information laws**RC charging**

Eq: $V_C(t) = V_0(1 - \exp(-t/RC))$

Law/use: Capacitor charge transient.

RC discharging

Eq: $V_C(t) = V_0 \exp(-t/RC)$

Law/use: Capacitor discharge transient.

RL current rise

Eq: $I(t) = I_0(1 - \exp(-tR/L))$

Law/use: Inductor current transient.

LC resonance

Eq: $\omega_0 = 1/\sqrt{LC}$

Law/use: Ideal resonant angular frequency.

RLC series impedance

Eq: $Z = R + j(\omega L - 1/(\omega C))$

Law/use: Frequency-domain impedance.

Capacitive reactance

Eq: $X_C = -1/(\omega C)$

Law/use: Imaginary impedance of capacitor.

Inductive reactance

Eq: $X_L = \omega L$

Law/use: Imaginary impedance of inductor.

AC real power

Eq: $P = V_{rms} I_{rms} \cos \phi$

Law/use: Average power in sinusoidal AC circuit.

Reactive power

Eq: $Q = V_{rms} I_{rms} \sin \phi$

Law/use: Oscillatory power in reactive elements.

Apparent power

Eq: $S = V_{rms} I_{rms}$

Law/use: Magnitude of complex power.

Power factor

Eq: $PF = \cos \phi = P/S$

Law/use: Fraction of apparent power doing real work.

Three phase power

Eq: $P = \sqrt{3} V_{LL} I_L \cos \phi$

Law/use: Balanced three-phase real power.

Star line voltage

Eq: $V_{LL} = \sqrt{3} V_{phase}$

Law/use: Wye voltage relation.

Delta current relation

Eq: $I_L = \sqrt{3} I_{phase}$

Law/use: Delta current relation.

Thevenin theorem

Eq: network $\rightarrow V_{th}$ in series with R_{th}

Law/use: Linear two-terminal network equivalence.

Norton theorem

Eq: network $\rightarrow I_N$ in parallel with R_N

Law/use: Current-source equivalent.

Maximum power transfer

Eq: $R_{load} = R_{source}$ for DC; $Z_L = Z_S^*$ for AC

Law/use: Load condition for maximum delivered power.

Superposition theorem circuits

Eq: response = sum response from each independent source

Law/use: Valid in linear circuits.

Tellegen theorem

Eq: $\sum_{branch} v_k i_k = 0$

Law/use: Power conservation in network topology.

Barkhausen criterion

Eq: $|A \beta| = 1$ and $\text{phase} = 0 \bmod 2\pi$

Law/use: Oscillation condition.

Nyquist stability criterion

Eq: encirclements of -1 determine closed-loop stability

Law/use: Frequency-domain stability law.

Bode gain phase relation

Eq: minimum-phase magnitude and phase linked by Hilbert transform

Law/use: Network response constraint.

Shockley diode equation

$$\text{Eq: } I = I_s (\exp(V_D / (n V_T)) - 1)$$

Law/use: Ideal diode current-voltage law.

Thermal voltage

$$\text{Eq: } V_T = kT/q$$

Law/use: Voltage scale for semiconductor junctions.

MOS square law saturation

$$\text{Eq: } I_D = 1/2 \mu C_{ox} W/L (V_{GS} - V_T)^2$$

Law/use: Long-channel MOSFET saturation approximation.

Johnson-Nyquist noise

$$\text{Eq: } v_n^2 = 4 kT R \Delta f$$

Law/use: Thermal noise voltage.

Shot noise

$$\text{Eq: } i_n^2 = 2 q I \Delta f$$

Law/use: Discrete charge carrier noise.

Friis transmission equation

$$\text{Eq: } P_r = P_t G_t G_r (\lambda / (4\pi R))^2$$

Law/use: Free-space radio link power.

Shannon-Hartley theorem

$$\text{Eq: } C = B \log_2(1 + S/N)$$

Law/use: Channel capacity under Gaussian noise.

Nyquist sampling theorem

$$\text{Eq: } f_s \geq 2 f_{\max}$$

Law/use: Minimum sampling frequency to avoid aliasing.

Telegrapher equations

$$\text{Eq: } dV/dx = -(R + j\omega L)I; \quad dI/dx = -(G + j\omega C)V$$

Law/use: Transmission-line voltage/current propagation.

Characteristic impedance

$$\text{Eq: } Z_0 = \sqrt{(R + j\omega L) / (G + j\omega C)}$$

Law/use: Transmission line impedance.

Reflection coefficient

$$\text{Eq: } \Gamma = (Z_L - Z_0) / (Z_L + Z_0)$$

Law/use: Voltage-wave reflection at load.

VSWR

$$\text{Eq: } VSWR = (1 + |\Gamma|) / (1 - |\Gamma|)$$

Law/use: Standing-wave ratio.

Transformer ideal law

$$\text{Eq: } V_s/V_p = N_s/N_p; \quad I_s/I_p = N_p/N_s$$

Law/use: Ideal transformer ratio.

Waves, sound, optics and radiation laws**Wave equation**

$$\text{Eq: } d^2y/dx^2 = (1/v^2) d^2y/dt^2$$

Law/use: Governs ideal wave propagation.

Wave speed string

$$\text{Eq: } v = \sqrt{T/\mu}$$

Law/use: Transverse wave speed on stretched string.

Sound speed ideal gas

$$\text{Eq: } v = \sqrt{\gamma R T/M}$$

Law/use: Adiabatic sound speed in gas.

Frequency wavelength

$$\text{Eq: } v = f \lambda$$

Law/use: Basic wave relation.

Angular frequency

$$\text{Eq: } \omega = 2\pi f$$

Law/use: Cycles per second to radians per second.

Wave number

$$\text{Eq: } k = 2\pi/\lambda$$

Law/use: Spatial angular frequency.

Phase velocity

$$\text{Eq: } v_p = \omega/k$$

Law/use: Speed of constant phase.

Group velocity

$$\text{Eq: } v_g = d\omega/dk$$

Law/use: Envelope/information speed in dispersive medium.

Doppler sound moving source

$$\text{Eq: } f_{\text{prime}} = f (v \pm v_o) / (v \mp v_s)$$

Law/use: Observed frequency shift for sound.

Relativistic Doppler

$$\text{Eq: } f_{\text{obs}} = f_{\text{src}} \sqrt{(1-\beta)/(1+\beta)} \text{ for receding}$$

Law/use: Light frequency shift from relative motion.

Standing wave string

$$\text{Eq: } f_n = n v / (2L)$$

Law/use: Both ends fixed harmonics.

Open pipe harmonics

Eq: $f_n = n v / (2L)$

Law/use: Open-open pipe resonances.

Closed pipe harmonics

Eq: $f_n = n v / (4L)$, odd n

Law/use: Open-closed pipe resonances.

Beat frequency

Eq: $f_{\text{beat}} = |f_1 - f_2|$

Law/use: Amplitude modulation between close frequencies.

Intensity inverse square

Eq: $I = P / (4\pi r^2)$

Law/use: Spherical wave intensity drop.

Decibel level

Eq: $\beta = 10 \log_{10}(I/I_0)$

Law/use: Logarithmic sound intensity level.

Acoustic impedance

Eq: $Z = \rho v$

Law/use: Pressure/particle velocity ratio.

Sabine reverberation

Eq: $T_{60} = 0.161 V/A$

Law/use: Room reverberation time estimate.

Huygens principle

Eq: each wavefront point acts as secondary source

Law/use: Wavefront propagation construction.

Fermat principle

Eq: travel time stationary

Law/use: Light path extremises optical time.

Law of reflection

Eq: $\theta_i = \theta_r$

Law/use: Angle of incidence equals angle of reflection.

Snell law

Eq: $n_1 \sin \theta_1 = n_2 \sin \theta_2$

Law/use: Refraction at interface.

Critical angle

Eq: $\sin \theta_c = n_2/n_1$ for $n_1 > n_2$

Law/use: Total internal reflection threshold.

Brewster law

Eq: $\tan \theta_B = n_2/n_1$

Law/use: Zero reflected p-polarized light.

Malus law

Eq: $I = I_0 \cos^2 \theta$

Law/use: Polarizer transmitted intensity.

Lens equation

Eq: $1/f = 1/d_o + 1/d_i$

Law/use: Thin lens imaging.

Magnification

Eq: $m = -d_i/d_o = h_i/h_o$

Law/use: Image magnification.

Mirror equation

Eq: $1/f = 1/d_o + 1/d_i$

Law/use: Spherical mirror relation.

Lensmaker equation

Eq: $1/f = (n-1)(1/R_1 - 1/R_2)$

Law/use: Thin lens focal length in air.

Young double slit

Eq: $d \sin \theta = m \lambda$

Law/use: Constructive interference maxima.

Single slit minima

Eq: $a \sin \theta = m \lambda$

Law/use: Diffraction minima.

Diffraction grating

Eq: $d \sin \theta = m \lambda$

Law/use: Grating maxima condition.

Bragg law

Eq: $n \lambda = 2 d \sin \theta$

Law/use: Crystal diffraction condition.

Rayleigh criterion

Eq: $\theta_{\text{min}} = 1.22 \lambda / D$

Law/use: Resolution limit for circular aperture.

Abbe limit

Eq: $d = \lambda / (2 \text{ NA})$

Law/use: Microscope diffraction resolution limit.

Beer-Lambert law

Eq: $A = \epsilon c l$

Law/use: Absorbance proportional to concentration and path length.

Lambert cosine law

Eq: $I(\theta) = I_0 \cos \theta$

Law/use: Radiant intensity from ideal diffuse surface.

Fresnel normal incidence

Eq: $R = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2$

Law/use: Reflectance at normal incidence.

Rayleigh scattering

Eq: intensity proportional $1/\lambda^4$

Law/use: Short wavelengths scatter more strongly.

Compton shift

Eq: $\Delta \lambda = \frac{h}{m_e c} (1 - \cos \theta)$

Law/use: Photon-electron scattering wavelength shift.

Photoelectric equation

Eq: $K_{\max} = h f - \phi$

Law/use: Maximum electron kinetic energy from photons.

Fluid mechanics, aerodynamics and hydrodynamic laws**Archimedes principle**

Eq: $F_b = \rho_{\text{fluid}} g V_{\text{displaced}}$

Law/use: Buoyant force equals weight of displaced fluid.

Pascal law

Eq: pressure change transmitted undiminished in enclosed fluid

Law/use: Foundation of hydraulics.

Hydrostatic pressure

Eq: $p = p_0 + \rho g h$

Law/use: Pressure increases with depth.

Continuity incompressible

Eq: $A_1 v_1 = A_2 v_2$

Law/use: Volume flow rate conservation.

Mass continuity

Eq: $\frac{d\rho}{dt} + \text{div}(\rho \mathbf{v}) = 0$

Law/use: Mass conservation in fluids.

Bernoulli equation

Eq: $p + \frac{1}{2} \rho v^2 + \rho g z = \text{constant}$

Law/use: Energy conservation along streamline.

Torricelli law

Eq: $v = \sqrt{2gh}$

Law/use: Efflux speed from hole under head h.

Navier-Stokes equation

Eq: $\rho \frac{D\mathbf{v}}{Dt} = -\text{grad } p + \mu \nabla^2 \mathbf{v} + \mathbf{f}$

Law/use: Momentum balance for viscous Newtonian fluid.

Euler fluid equation

Eq: $\rho \frac{D\mathbf{v}}{Dt} = -\text{grad } p + \mathbf{f}$

Law/use: Inviscid flow momentum equation.

Newton viscosity law

Eq: $\tau = \mu \frac{du}{dy}$

Law/use: Shear stress proportional to velocity gradient.

Poiseuille law

Eq: $Q = \frac{\pi r^4 \Delta p}{8 \mu L}$

Law/use: Laminar flow rate through circular pipe.

Darcy law

Eq: $Q = -\frac{k A \Delta p}{\mu L}$

Law/use: Flow through porous media.

Darcy-Weisbach

Eq: $\Delta p = f_D \left(\frac{L}{D} \right) \rho v^2 / 2$

Law/use: Pipe friction pressure loss.

Reynolds number

Eq: $Re = \rho v L / \mu$

Law/use: Ratio of inertial to viscous forces.

Froude number

Eq: $Fr = v / \sqrt{gL}$

Law/use: Ratio of flow speed to gravity wave speed.

Mach number

Eq: $Ma = v / c_s$

Law/use: Flow speed relative to sound.

Weber number

Eq: $We = \rho v^2 L / \sigma$

Law/use: Inertial to surface tension forces.

Bond number

Eq: $Bo = \rho g L^2 / \sigma$

Law/use: Gravity to surface tension forces.

Capillary number

Eq: $Ca = \mu v / \sigma$

Law/use: Viscous to surface tension forces.

Prandtl number

Eq: $Pr = \nu / \alpha$

Law/use: Momentum diffusivity to thermal diffusivity.

Nusselt number

Eq: $Nu = hL/k$

Law/use: Convective to conductive heat transfer.

Rayleigh number

Eq: $Ra = g \beta \Delta T L^3 / (\nu \alpha)$

Law/use: Natural convection criterion.

Grashof number

Eq: $Gr = g \beta \Delta T L^3 / \nu^2$

Law/use: Buoyancy to viscous forces.

Strouhal number

Eq: $St = fL/v$

Law/use: Oscillating flow scaling.

Stokes law

Eq: $F_d = 6 \pi \mu r v$

Law/use: Creeping-flow drag on sphere.

Stokes settling velocity

Eq: $v_t = 2 r^2 (\rho_p - \rho_f) g / (9 \mu)$

Law/use: Terminal speed of small sphere.

Young-Laplace equation

Eq: $\Delta p = \gamma (1/R_1 + 1/R_2)$

Law/use: Pressure jump across curved interface.

Jurin capillary rise

Eq: $h = 2 \gamma \cos \theta / (\rho g r)$

Law/use: Capillary rise in tube.

Kutta-Joukowski lift

Eq: $L_{\text{prime}} = \rho v \Gamma$

Law/use: Lift per span from circulation.

Magnus effect

Eq: lift from rotating body in flow

Law/use: Spin-induced transverse force.

Venturi effect

Eq: higher speed -> lower pressure

Law/use: Consequence of Bernoulli.

Coanda effect

Eq: jet attaches to nearby curved surface

Law/use: Fluid entrainment and pressure effect.

Kelvin circulation theorem

Eq: $D \Gamma / Dt = 0$ for inviscid barotropic conservative flow

Law/use: Circulation conservation.

Helmholtz vortex theorem

Eq: vortex lines move with ideal fluid

Law/use: Ideal vortex dynamics.

Rankine-Hugoniot conditions

Eq: conservation jumps across shock

Law/use: Shock discontinuity relations.

Relativity, spacetime and cosmology equations**Special relativity postulate 1**

Eq: laws of physics same in all inertial frames

Law/use: Principle of relativity.

Special relativity postulate 2

Eq: c same for all inertial observers

Law/use: Constancy of light speed.

Lorentz factor

Eq: $\gamma = 1/\sqrt{1-v^2/c^2}$

Law/use: Relativistic scaling factor.

Time dilation

Eq: $\Delta t = \gamma \Delta \tau$

Law/use: Moving clocks appear slower.

Length contraction

Eq: $L = L_0/\gamma$

Law/use: Moving lengths contract along motion.

Relativistic velocity addition

Eq: $u_{\text{prime}} = (u-v)/(1-uv/c^2)$

Law/use: Velocity transformation.

Mass-energy equivalence

Eq: $E_0 = m c^2$

Law/use: Rest energy of mass.

Energy-momentum relation

Eq: $E^2 = (pc)^2 + (mc^2)^2$

Law/use: Relativistic energy and momentum.

Relativistic momentum

Eq: $p = \gamma m v$

Law/use: Momentum at high speed.

Minkowski interval

Eq: $s^2 = c^2 t^2 - x^2 - y^2 - z^2$

Law/use: Invariant spacetime interval.

Four-velocity norm

Eq: $U_\mu U^\mu = c^2$

Law/use: Invariant norm of four-velocity.

Four-momentum norm

Eq: $P_\mu P^\mu = m^2 c^2$

Law/use: Invariant mass-shell condition.

Equivalence principle

Eq: local gravity indistinguishable from acceleration

Law/use: Foundation of general relativity.

Einstein field equation

Eq: $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$

Law/use: Spacetime curvature sourced by stress-energy.

Geodesic equation

Eq: $d^2x^\mu/d\tau^2 + \Gamma^\mu_{\alpha\beta} dx^\alpha/d\tau dx^\beta/d\tau = 0$

Law/use: Free-fall motion in curved spacetime.

Stress-energy conservation

Eq: $\nabla_\mu T^{\mu\nu} = 0$

Law/use: Local energy-momentum conservation in GR.

Schwarzschild radius

Eq: $r_s = 2GM/c^2$

Law/use: Event horizon radius for non-rotating black hole.

Gravitational time dilation

Eq: $dt_{\text{far}} = dt_{\text{local}}/\sqrt{1-2GM/(rc^2)}$

Law/use: Clocks deeper in gravity run slower.

Gravitational redshift

Eq: $z \approx \Delta\Phi/c^2$

Law/use: Photon loses frequency escaping gravity.

Light deflection weak field

Eq: $\alpha = 4GM/(c^2 b)$

Law/use: GR bending angle by mass.

Perihelion precession

Eq: $\Delta\phi = 6\pi GM/(a(1-e^2)c^2)$

Law/use: GR correction per orbit.

Friedmann equation

Eq: $(\dot{a}/a)^2 = 8\pi G \rho/3 - k c^2/a^2 + \Lambda c^2/3$

Law/use: Cosmic scale-factor evolution.

Acceleration Friedmann

Eq: $\ddot{a}/a = -4\pi G(\rho+3p/c^2)/3 + \Lambda c^2/3$

Law/use: Cosmic acceleration equation.

Hubble-Lemaître law

Eq: $v = H_0 d$

Law/use: Recession velocity proportional to distance.

Bekenstein-Hawking entropy

Eq: $S = k c^3 A/(4 G \hbar)$

Law/use: Black hole entropy proportional to horizon area.

Hawking temperature

Eq: $T_H = \hbar c^3/(8\pi G M k_B)$

Law/use: Black hole thermal temperature.

Black hole area theorem

Eq: $dA \geq 0$ classically

Law/use: Horizon area non-decreasing in classical GR.

No-hair theorem

Eq: stationary black holes described by M,J,Q

Law/use: Black holes have limited external parameters.

Shapiro delay

Eq: extra light travel time near mass

Law/use: GR time delay of signals passing near gravity.

Quantum mechanics, atomic laws and quantum information**Planck quantum relation**

Eq: $E = h f = \hbar \omega$

Law/use: Energy of photon/quantum proportional to frequency.

de Broglie wavelength

Eq: $\lambda = h/p$

Law/use: Matter-wave wavelength.

Schrodinger time dependent

Eq: $i \hbar \frac{d\psi}{dt} = H \psi$

Law/use: Quantum state time evolution.

Schrodinger time independent

Eq: $H \psi = E \psi$

Law/use: Energy eigenvalue equation.

Born rule

Eq: $P(x) = |\psi(x)|^2$

Law/use: Probability density from wavefunction amplitude.

Normalization

Eq: $\int |\psi|^2 dx = 1$

Law/use: Total probability is one.

Heisenberg uncertainty

Eq: $\Delta x \Delta p \geq \hbar/2$

Law/use: Position-momentum uncertainty bound.

Energy-time uncertainty

Eq: $\Delta E \Delta t \geq \hbar/2$

Law/use: Energy-time spread relation.

Commutation canonical

Eq: $[x, p] = i \hbar$

Law/use: Canonical quantum commutator.

Angular momentum commutator

Eq: $[L_x, L_y] = i \hbar L_z$

Law/use: Rotational quantum algebra.

Spin quantization

Eq: $S^2 = s(s+1)\hbar^2$; $S_z = m_s \hbar$

Law/use: Spin magnitude/projection eigenvalues.

Pauli exclusion principle

Eq: no two identical fermions occupy same quantum state

Law/use: Core law for electronic structure.

Superposition principle quantum

Eq: $\psi = c_1 \psi_1 + c_2 \psi_2$

Law/use: Linear combinations of states are states.

Projection postulate

Eq: measurement projects state onto eigenstate

Law/use: Ideal measurement update rule.

Ehrenfest theorem

Eq: $\frac{d\langle x \rangle}{dt} = \langle p \rangle / m$; $\frac{d\langle p \rangle}{dt} = -\langle \text{grad } V \rangle$

Law/use: Quantum averages obey classical-like equations.

Fermi golden rule

Eq: transition rate = $(2\pi/\hbar) |M|^2 \rho(E)$

Law/use: Perturbative transition rate.

Quantum tunnelling WKB

Eq: $T \sim \exp(-2 \int \kappa dx)$

Law/use: Barrier penetration probability.

Harmonic oscillator energy

Eq: $E_n = \hbar \omega (n + 1/2)$

Law/use: Quantized oscillator levels.

Particle in box

Eq: $E_n = n^2 \hbar^2 / (8mL^2)$

Law/use: Energy levels in 1D infinite well.

Hydrogen energy levels

Eq: $E_n = -13.6 \text{ eV} / n^2$

Law/use: Bohr/Schrodinger hydrogen spectrum.

Bohr angular momentum

Eq: $m v r = n \hbar$

Law/use: Old quantum quantization rule.

Bohr radius

Eq: $a_0 = 4\pi \epsilon_0 \hbar^2 / (m_e e^2)$

Law/use: Hydrogen ground-state length scale.

Rydberg formula

Eq: $1/\lambda = R(1/n_1^2 - 1/n_2^2)$

Law/use: Hydrogen spectral lines.

Dirac equation

Eq: $(i \hbar \gamma^\mu \partial_\mu - mc) \psi = 0$

Law/use: Relativistic spin-1/2 wave equation.

Klein-Gordon equation

Eq: $(\Box + m^2 \hbar^2 / c^2) \phi = 0$

Law/use: Relativistic scalar wave equation.

Pauli equation

Eq: spin magnetic coupling included in Schrodinger equation

Law/use: Nonrelativistic spin-1/2 particle in EM field.

Bose-Einstein distributionEq: $n(E) = 1 / (\exp((E - \mu) / kT) - 1)$

Law/use: Boson occupation number.

Fermi-Dirac distributionEq: $n(E) = 1 / (\exp((E - \mu) / kT) + 1)$

Law/use: Fermion occupation number.

Maxwell-Boltzmann limitEq: $n(E)$ proportional $\exp(-(E - \mu) / kT)$

Law/use: Classical low-density distribution.

Spin-statistics theorem

Eq: integer spin bosons; half-integer spin fermions

Law/use: Relativistic QFT connection.

CPT theorem

Eq: combined charge-parity-time symmetry conserved in local Lorentz QFT

Law/use: Deep symmetry theorem.

Bell theorem

Eq: local hidden variables cannot reproduce all quantum predictions

Law/use: Entanglement nonclassicality.

Bell CHSH inequalityEq: $|S| \leq 2$ for local hidden variables

Law/use: Experimentally testable inequality.

No-cloning theorem

Eq: unknown arbitrary quantum state cannot be copied

Law/use: Linearity forbids perfect cloning.

No-deleting theorem

Eq: unknown quantum information cannot be deleted unitarily

Law/use: Information conservation property.

No-broadcasting theorem

Eq: noncommuting states cannot be broadcast

Law/use: Mixed-state generalisation of no-cloning.

Adiabatic theorem

Eq: slow Hamiltonian change keeps system in instantaneous eigenstate

Law/use: Basis of adiabatic quantum computing.

Quantum Zeno effect

Eq: frequent measurement inhibits evolution

Law/use: Observation changes transition dynamics.

Density matrix evolutionEq: $d\rho/dt = -(i/\hbar)[H, \rho]$

Law/use: Closed quantum system state evolution.

Lindblad equationEq: $d\rho/dt = -i[H, \rho]/\hbar + \sum L \rho L^\dagger - 1/2\{L^\dagger L, \rho\}$

Law/use: Markovian open quantum system law.

Path integral amplitudeEq: $K = \int \exp(iS/\hbar) D[\text{path}]$

Law/use: Sum-over-histories formulation.

Aharonov-Bohm phaseEq: $\Delta\phi = q/\hbar \int A \cdot dl$

Law/use: Potential affects quantum phase.

Casimir force platesEq: $F/A = -\pi^2 \hbar c / (240 a^4)$

Law/use: Vacuum fluctuation attraction between plates.

Lamb shift

Eq: QED radiative correction to hydrogen levels

Law/use: Vacuum fluctuations shift atomic energy.

Zeeman splittingEq: $\Delta E = \mu_B g m_j B$

Law/use: Magnetic field splits energy levels.

Stark effect

Eq: energy shift from electric field

Law/use: Electric field splits/shifts atomic levels.

Selection rule dipoleEq: $\Delta l = \pm 1$; $\Delta m = 0, \pm 1$

Law/use: Allowed electric dipole transitions.

Hund first rule

Eq: maximum spin multiplicity lowest energy

Law/use: Atomic electron configuration rule.

Aufbau principle

Eq: electrons fill lower energy orbitals first

Law/use: Orbital filling rule.

Franck-Condon principle

Eq: electronic transitions vertical in nuclear coordinate

Law/use: Vibronic transition intensity rule.

Bloch theorem

$$\text{Eq: } \psi_k(r) = u_k(r) \exp(i k \cdot r)$$

Law/use: Electron states in periodic crystal potential.

Band gap

$$\text{Eq: } E_g = E_{\text{conduction_min}} - E_{\text{valence_max}}$$

Law/use: Energy gap between bands.

Quantum Hall conductance

$$\text{Eq: } G_{xy} = \nu e^2/h$$

Law/use: Quantized transverse conductance.

Landau levels

$$\text{Eq: } E_n = \hbar \omega_c (n + 1/2)$$

Law/use: Charged particles in magnetic field.

BCS gap relation

$$\text{Eq: } \Delta(0) = 1.76 k_B T_c$$

Law/use: Weak-coupling superconducting gap.

Nuclear physics, particle physics and quantum field laws**Radioactive decay law**

$$\text{Eq: } N(t) = N_0 \exp(-\lambda t)$$

Law/use: Number of undecayed nuclei decreases exponentially.

Activity

$$\text{Eq: } A = \lambda N$$

Law/use: Decay rate.

Half life

$$\text{Eq: } t_{1/2} = \ln 2 / \lambda$$

Law/use: Time for half nuclei to decay.

Mean lifetime

$$\text{Eq: } \tau = 1/\lambda$$

Law/use: Average survival time.

Bateman equations

Eq: decay chain populations from coupled first-order equations

Law/use: Sequential radioactive decay.

Binding energy

$$\text{Eq: } B = (\Delta m) c^2$$

Law/use: Mass defect converted to nuclear binding energy.

Semi empirical mass formula

$$\text{Eq: } B = a_v A - a_s A^{2/3} - a_c \frac{Z(Z-1)}{A^{1/3}} - a_a \frac{(A-2Z)^2}{A} \pm \delta$$

Law/use: Liquid-drop nuclear binding estimate.

Rutherford scattering

$$\text{Eq: } d\sigma/d\Omega \propto 1/\sin^4(\theta/2)$$

Law/use: Coulomb scattering angular distribution.

Geiger-Nuttall law

$$\text{Eq: } \log t_{1/2} = a \frac{Z}{\sqrt{E_\alpha}} + b$$

Law/use: Alpha decay half-life relationship.

Fission chain condition

$$\text{Eq: } k_{\text{eff}} > 1 \text{ supercritical; } = 1 \text{ critical; } < 1 \text{ subcritical}$$

Law/use: Neutron multiplication criterion.

Critical mass principle

Eq: minimum mass/geometry for sustained chain reaction

Law/use: Nuclear reactor/weapon physics condition.

Lawson criterion

Eq: $n T \tau_E$ sufficient for fusion ignition

Law/use: Fusion plasma confinement condition.

Fusion triple product

$$\text{Eq: } n T \tau_E$$

Law/use: Performance measure for fusion.

Q value reaction

$$\text{Eq: } Q = (m_{\text{initial}} - m_{\text{final}}) c^2$$

Law/use: Energy release/absorption in nuclear reaction.

Bethe-Bloch formula

$$\text{Eq: } -dE/dx = \text{stopping power from ionization interactions}$$

Law/use: Charged particle energy loss in matter.

Breit-Wigner resonance

$$\text{Eq: } \sigma(E) \propto \frac{\Gamma^2}{(E - E_0)^2 + \Gamma^2/4}$$

Law/use: Resonance cross-section shape.

Neutrino oscillation length

$$\text{Eq: } L_{\text{osc}} = 4\pi E / \Delta m^2$$

Law/use: Flavor oscillation scale in natural units.

Standard Model gauge group

$$\text{Eq: } SU(3)_C \times SU(2)_L \times U(1)_Y$$

Law/use: Gauge symmetry of known particle physics.

QED coupling

$$\text{Eq: } \alpha = e^2 / (4\pi \epsilon_0 \hbar c)$$

Law/use: Fine-structure constant.

QCD asymptotic freedomEq: α_s decreases at high energy

Law/use: Quarks interact more weakly at short distances.

Confinement principle

Eq: isolated color charges not observed

Law/use: Quarks/gluons confined in hadrons.

Higgs mechanism

Eq: gauge bosons and fermions acquire mass through Higgs field interactions

Law/use: Electroweak symmetry breaking.

Goldstone theoremEq: spontaneous continuous symmetry breaking $\rightarrow$ massless modes

Law/use: Symmetry-breaking theorem.

Ward identity

Eq: gauge symmetry constrains correlation functions

Law/use: Quantum gauge conservation law.

Optical theorem

Eq: Im forward scattering amplitude relates to total cross section

Law/use: Unitarity of scattering.

Solid state, materials science and condensed matter laws**Stress**Eq: $\sigma = F/A$

Law/use: Force per area in material.

StrainEq: $\epsilon = \Delta L/L$

Law/use: Fractional deformation.

Young modulusEq: $\sigma = E \epsilon$

Law/use: Linear elastic axial stress-strain law.

Shear modulusEq: $\tau = G \gamma$

Law/use: Linear shear stress-strain law.

Bulk modulusEq: $B = -V dp/dV$

Law/use: Resistance to compression.

Poisson ratioEq: $\nu = -\epsilon_{\text{trans}}/\epsilon_{\text{axial}}$

Law/use: Transverse contraction ratio.

Hooke generalizedEq: $\sigma_{ij} = C_{ijkl} \epsilon_{kl}$

Law/use: Linear elasticity tensor form.

Hertz contactEq: contact force proportional $\delta^{3/2}$

Law/use: Elastic sphere contact law.

Griffith fracture criterionEq: $\sigma_c = \sqrt{2E \gamma / (\pi a)}$

Law/use: Crack growth energy criterion.

Paris law fatigueEq: $da/dN = C(\Delta K)^m$

Law/use: Fatigue crack growth per cycle.

Basquin lawEq: $\sigma_a = \sigma_f (2N_f)^b$

Law/use: High-cycle fatigue stress-life relation.

Coffin-Manson lawEq: $\epsilon_p/2 = \epsilon_f (2N_f)^c$

Law/use: Low-cycle fatigue strain-life relation.

Miner ruleEq: $\sum n_i/N_i = 1$

Law/use: Cumulative fatigue damage approximation.

Hall-Petch relationEq: $\sigma_y = \sigma_0 + k d^{-1/2}$

Law/use: Smaller grains raise yield strength.

von Mises criterionEq: $\sigma_{vm} \geq \sigma_y$

Law/use: Ductile yielding criterion.

Tresca criterionEq: max shear stress reaches $\text{yield}/2$

Law/use: Maximum shear stress yield criterion.

Mohr-Coulomb criterionEq: $\tau = c + \sigma_n \tan \phi$

Law/use: Shear failure in soils/rocks.

Fick first lawEq: $J = -D \text{grad } c$

Law/use: Diffusive flux down concentration gradient.

Fick second law

Eq: $dc/dt = D \nabla^2 c$

Law/use: Diffusion equation.

Arrhenius equation

Eq: $k = A \exp(-E_a/(RT))$

Law/use: Temperature dependence of reaction or diffusion rates.

Nernst-Einstein relation

Eq: $D = \mu kT/q$

Law/use: Diffusion coefficient and mobility link.

Matthiessen rule

Eq: $\rho_{\text{total}} = \rho_{\text{thermal}} + \rho_{\text{impurity}} + \dots$

Law/use: Resistivity contributions add approximately.

Wiedemann-Franz law

Eq: $\kappa/(\sigma T) = L$

Law/use: Thermal/electrical conductivity ratio in metals.

Bragg law

Eq: $n \lambda = 2d \sin \theta$

Law/use: Crystal diffraction.

Scherrer equation

Eq: $\tau = K \lambda / (\beta \cos \theta)$

Law/use: Crystallite size from diffraction broadening.

Vegard law

Eq: lattice parameter varies linearly with alloy composition

Law/use: Approximate alloy lattice relation.

Hume-Rothery rules

Eq: solid solubility favoured by size, structure, valence, electronegativity similarity

Law/use: Alloy formation rules.

Stoner criterion

Eq: $I D(E_F) > 1$

Law/use: Itinerant ferromagnetism condition.

Curie temperature law

Eq: magnetic ordering lost above T_C

Law/use: Ferromagnetic transition.

Nernst equation solid electrochem

Eq: $E = E^0 - (RT/nF) \ln Q$

Law/use: Electrochemical equilibrium potential.

Plasma physics and magnetohydrodynamic laws**Plasma frequency**

Eq: $\omega_p = \sqrt{n e^2 / (\epsilon_0 m_e)}$

Law/use: Natural electron oscillation frequency.

Debye length

Eq: $\lambda_D = \sqrt{\epsilon_0 kT / (n e^2)}$

Law/use: Charge shielding length in plasma.

Debye shielding

Eq: $\phi(r)$ proportional $\exp(-r/\lambda_D)/r$

Law/use: Electrostatic potentials are screened.

Gyrofrequency

Eq: $\omega_c = qB/m$

Law/use: Charged particle rotation frequency.

Gyroradius

Eq: $r_L = m v_{\perp} / (qB)$

Law/use: Larmor radius.

Alfven speed

Eq: $v_A = B / \sqrt{\mu_0 \rho}$

Law/use: MHD wave speed along magnetic field.

Magnetic Reynolds number

Eq: $R_m = \mu_0 \sigma v L$

Law/use: Advection vs diffusion of magnetic field.

Frozen-in flux theorem

Eq: magnetic flux moves with ideal conducting fluid

Law/use: Ideal MHD magnetic field behaviour.

MHD induction equation

Eq: $\partial B / \partial t = \text{curl}(v \times B) + \eta \nabla^2 B$

Law/use: Magnetic field evolution in conducting fluid.

Vlasov equation

Eq: $df/dt + v \cdot \text{grad}_x f + (F/m) \cdot \text{grad}_v f = 0$

Law/use: Collisionless plasma distribution evolution.

Bohm diffusion

Eq: $D_B \sim kT / (16 e B)$

Law/use: Empirical cross-field diffusion estimate.

Spitzer resistivity

Eq: η proportional $T_e^{-3/2}$

Law/use: Classical plasma resistivity scaling.

Paschen law

Eq: breakdown voltage depends on $p \cdot d$
 Law/use: Gas breakdown condition.

Townsend avalanche

Eq: ionization grows exponentially with distance
 Law/use: Electrical discharge multiplication.

Child-Langmuir law

Eq: $J = (4/9) \epsilon_0 \sqrt{2e/m} V^{3/2}/d^2$
 Law/use: Space-charge-limited current.

Kruskal-Shafranov limit

Eq: safety factor $q > 1$ for kink stability
 Law/use: Tokamak plasma stability condition.

Landau damping

Eq: wave damping without collisions by resonant particles
 Law/use: Kinetic plasma damping.

Magnetic mirror condition

Eq: $\sin^2 \alpha_0 \geq B_0/B_{\max}$ for reflection
 Law/use: Particle trapping condition.

Bennett pinch relation

Eq: magnetic pressure balances plasma pressure
 Law/use: Pinch equilibrium.

Saha ionization equation

Eq: ionization balance by temperature and density
 Law/use: Thermal ionization law.

Astrophysics and cosmology laws**Stefan-Boltzmann luminosity**

Eq: $L = 4\pi R^2 \sigma T^4$
 Law/use: Star luminosity from radius and temperature.

Wien law astronomy

Eq: $\lambda_{\max} = b/T$
 Law/use: Stellar colour-temperature relation.

Distance modulus

Eq: $m - M = 5 \log_{10}(d/10 \text{ pc})$
 Law/use: Magnitude-distance relation.

Inverse square flux

Eq: $F = L/(4\pi d^2)$
 Law/use: Observed flux from luminosity.

Tully-Fisher relation

Eq: L proportional v_{rot}^α
 Law/use: Spiral galaxy luminosity-rotation relation.

Faber-Jackson relation

Eq: L proportional σ^4
 Law/use: Elliptical galaxy luminosity-dispersion relation.

Mass-luminosity relation

Eq: L proportional M^3 to M^4 approximately
 Law/use: Main-sequence star scaling.

Hydrostatic equilibrium star

Eq: $dp/dr = -G M(r) \rho/r^2$
 Law/use: Pressure gradient balances gravity.

Mass continuity star

Eq: $dM/dr = 4\pi r^2 \rho$
 Law/use: Stellar mass accumulation equation.

Radiative transport

Eq: $dT/dr = -3 \kappa \rho L/(16 \pi a c T^3 r^2)$
 Law/use: Radiative temperature gradient.

Lane-Emden equation

Eq: $1/\xi^2 d/d\xi(\xi^2 d\theta/d\xi) = -\theta^n$
 Law/use: Polytropic stellar structure.

Eddington luminosity

Eq: $L_{\text{Edd}} = 4\pi G M c/\kappa$
 Law/use: Radiation pressure gravity balance.

Bondi accretion

Eq: $\dot{M} \sim 4\pi \lambda (GM)^2 \rho/c_s^3$
 Law/use: Spherical accretion rate.

Roche lobe Eggleton

Eq: $R_L/a = 0.49q^{2/3}/(0.6q^{2/3} + \ln(1+q^{1/3}))$
 Law/use: Binary-star Roche lobe approximation.

Parker solar wind

Eq: transonic outflow from hot corona
 Law/use: Solar wind model.

Toomre criterion

Eq: $Q = \kappa \sigma/(\pi G \Sigma) > 1$ stable
 Law/use: Disk gravitational stability.

Kennicutt-Schmidt lawEq: $\Sigma_{\text{SFR}} \propto \Sigma_{\text{gas}}^N$

Law/use: Galaxy star-formation relation.

Salpeter IMFEq: $dN/dM \propto M^{-2.35}$

Law/use: Initial mass function estimate.

Sachs-Wolfe effectEq: $\Delta T/T \sim \Phi/(3c^2)$

Law/use: CMB gravitational temperature anisotropy.

Inflation slow rollEq: $\epsilon \ll 1; |\eta| \ll 1$

Law/use: Inflation condition.

Physical chemistry laws used in physics**Conservation of mass chemistry**

Eq: mass reactants = mass products

Law/use: Classical chemical reaction mass conservation.

Definite proportions

Eq: compound has fixed mass ratio of elements

Law/use: Stoichiometric composition law.

Multiple proportions

Eq: element mass ratios are small whole numbers

Law/use: Atomic theory evidence.

Law of mass action

Eq: rate/product activities proportional to reactant activities powers

Law/use: Reaction equilibrium/kinetics basis.

Equilibrium constantEq: $K = \frac{\text{product } a_{\text{products}}^{\nu}}{\text{product } a_{\text{reactants}}^{\nu}}$

Law/use: Mass action equilibrium expression.

Reaction Gibbs relationEq: $\Delta G = \Delta G^0 + RT \ln Q$

Law/use: Free energy and reaction quotient.

Nernst equationEq: $E = E^0 - (RT/nF) \ln Q$

Law/use: Electrochemical cell potential.

Faraday electrolysis first lawEq: $m \propto Q$

Law/use: Mass deposited proportional to charge.

Faraday electrolysis second lawEq: $m \propto \text{equivalent weight}$

Law/use: Mass deposited depends on molar mass and valence.

Hess lawEq: ΔH total independent of path

Law/use: Enthalpy is state function.

Kirchhoff thermochemistryEq: $d(\Delta H)/dT = \Delta C_p$

Law/use: Reaction enthalpy temperature dependence.

Le Chatelier principle

Eq: equilibrium shifts to oppose imposed change

Law/use: Qualitative equilibrium response.

Van t Hoff equationEq: $d \ln K/dT = \Delta H^0/(RT^2)$

Law/use: Equilibrium constant temperature dependence.

Henderson-HasselbalchEq: $\text{pH} = \text{pK}_a + \log_{10}(A^-/HA)$

Law/use: Buffer pH equation.

Michaelis-MentenEq: $v = V_{\text{max}} [S]/(K_M + [S])$

Law/use: Enzyme saturation kinetics.

Rate law first orderEq: $[A] = [A]^0 \exp(-kt)$

Law/use: First-order decay/conversion.

Rate law second orderEq: $1/[A] = 1/[A]^0 + kt$

Law/use: Second-order single reactant integrated law.

Arrhenius lawEq: $k = A \exp(-E_a/RT)$

Law/use: Reaction rate temperature dependence.

Eyring equationEq: $k = (k_B T/h) \exp(-\Delta G_{\text{dagger}}/RT)$

Law/use: Transition state theory rate.

Debye-Huckel limiting lawEq: $\log \gamma_{\pm} = -A z_{\pm}^2 \sqrt{I}$

Law/use: Activity coefficient in dilute electrolyte.

Kohlrausch law

Eq: $\Lambda_m = \Lambda_m^0 - K \sqrt{c}$

Law/use: Molar conductivity concentration law.

Ostwald dilution law

Eq: $K_a = c \alpha^2 / (1 - \alpha)$

Law/use: Weak electrolyte dissociation.

Huckel rule

Eq: $4n+2$ pi electrons aromatic

Law/use: Aromatic stability rule.

Woodward-Hoffmann rules

Eq: orbital symmetry controls pericyclic reactions

Law/use: Conservation of orbital symmetry.

Mathematical laws and theorems used in physics**Pythagorean theorem**

Eq: $a^2 + b^2 = c^2$

Law/use: Right-triangle geometry law.

Law of sines

Eq: $a/\sin A = b/\sin B = c/\sin C$

Law/use: Triangle side-angle relation.

Law of cosines

Eq: $c^2 = a^2 + b^2 - 2ab \cos C$

Law/use: General triangle metric law.

Euler formula

Eq: $e^{i\theta} = \cos \theta + i \sin \theta$

Law/use: Complex exponential relation.

Euler identity

Eq: $e^{i\pi} + 1 = 0$

Law/use: Links e, i, pi, 1 and 0.

Derivative definition

Eq: $f'(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)]/h$

Law/use: Instantaneous rate of change.

Fundamental theorem calculus

Eq: $d/dx \int_a^x f(t) dt = f(x)$

Law/use: Differentiation and integration are inverse.

Integration by parts

Eq: $\int u dv = uv - \int v du$

Law/use: Product rule integrated.

Chain rule

Eq: $d f(g(x))/dx = f'(g(x)) g'(x)$

Law/use: Derivative of composite function.

Product rule

Eq: $(fg)' = f'g + fg'$

Law/use: Derivative of product.

Taylor theorem

Eq: $f(x) = \sum f^{(n)}(a)(x-a)^n/n!$

Law/use: Local power series approximation.

Gradient

Eq: $\text{grad } f = \text{vector of partial derivatives}$

Law/use: Direction of steepest increase.

Divergence theorem

Eq: $\int_V \text{div } F dV = \oint_S F \cdot dA$

Law/use: Flux-volume relation.

Stokes theorem

Eq: $\int_S \text{curl } F \cdot dA = \oint_{\text{boundary}} F \cdot dl$

Law/use: Curl-circulation relation.

Green theorem

Eq: $\oint (P dx + Q dy) = \int (dQ/dx - dP/dy) dA$

Law/use: Planar Stokes theorem.

Laplacian

Eq: $\nabla^2 f = \text{div grad } f$

Law/use: Diffusion/wave/potential operator.

Fourier transform

Eq: $F(k) = \int f(x) \exp(-ikx) dx$

Law/use: Frequency-domain representation.

Inverse Fourier transform

Eq: $f(x) = 1/(2\pi) \int F(k) \exp(ikx) dk$

Law/use: Recover signal from spectrum.

Convolution theorem

Eq: $FT(f * g) = FT(f) FT(g)$

Law/use: Convolution becomes multiplication.

Parseval theorem

Eq: $\int |f|^2 dx = 1/(2\pi) \int |F|^2 dk$

Law/use: Energy conserved between domains.

Laplace transform

Eq: $F(s) = \int_0^\infty f(t) \exp(-st) dt$

Law/use: Complex frequency transform.

Central limit theorem

Eq: sum of many independent variables -> normal distribution

Law/use: Foundation of statistical physics/errors.

Law large numbers

Eq: sample mean -> expected value

Law/use: Average converges with many trials.

Bayes theorem

Eq: $P(A|B) = P(B|A)P(A)/P(B)$

Law/use: Probability update law.

Cauchy-Schwarz inequality

Eq: $|\langle u, v \rangle| \leq \|u\| \|v\|$

Law/use: Inner product bound.

Jensen inequality

Eq: $f(E[X]) \leq E[f(X)]$ for convex f

Law/use: Convexity expectation law.

Eigenvalue equation

Eq: $A v = \lambda v$

Law/use: Modes of linear transformation.

Determinant area scalingEq: $\det(A)$ gives oriented volume scale

Law/use: Linear map volume law.

Gaussian integral

Eq: $\int_{-\infty}^{\infty} \exp(-a x^2) dx = \sqrt{\pi/a}$

Law/use: Core normalisation integral.

Dirac delta property

Eq: $\int f(x) \delta(x-a) dx = f(a)$

Law/use: Sampling distribution.

Euler-Lagrange field

Eq: $dL/d\phi - d_\mu(dL/d(d_\mu \phi)) = 0$

Law/use: Field equation from action.

Mechanics formula appendix - kinematics, rotation and rigid bodies**Moment of inertia - point mass**

Eq: $I = m r^2$

Law/use: Rigid body rotational inertia formula.

Moment of inertia - solid disk/cylinder axis

Eq: $I = \frac{1}{2} M R^2$

Law/use: Rigid body rotational inertia formula.

Moment of inertia - thin hoop axis

Eq: $I = M R^2$

Law/use: Rigid body rotational inertia formula.

Moment of inertia - solid sphere

Eq: $I = \frac{2}{5} M R^2$

Law/use: Rigid body rotational inertia formula.

Moment of inertia - spherical shell

Eq: $I = \frac{2}{3} M R^2$

Law/use: Rigid body rotational inertia formula.

Moment of inertia - rod centre

Eq: $I = \frac{1}{12} M L^2$

Law/use: Rigid body rotational inertia formula.

Moment of inertia - rod end

Eq: $I = \frac{1}{3} M L^2$

Law/use: Rigid body rotational inertia formula.

Moment of inertia - rectangular plate centre

Eq: $I = \frac{1}{12} M (a^2 + b^2)$

Law/use: Rigid body rotational inertia formula.

Kinematics - constant acceleration velocity

Eq: $v = v_0 + a t$

Law/use: Constant-acceleration or rotational kinematic relation.

Kinematics - constant acceleration position

Eq: $x = x_0 + v_0 t + \frac{1}{2} a t^2$

Law/use: Constant-acceleration or rotational kinematic relation.

Kinematics - velocity displacement

Eq: $v^2 = v_0^2 + 2a(x - x_0)$

Law/use: Constant-acceleration or rotational kinematic relation.

Kinematics - average velocity constant a

Eq: $x - x_0 = \frac{1}{2}(v + v_0)t$

Law/use: Constant-acceleration or rotational kinematic relation.

Kinematics - angular velocity

Eq: $\omega = \omega_0 + \alpha t$

Law/use: Constant-acceleration or rotational kinematic relation.

Kinematics - angular displacement

Eq: $\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$
 Law/use: Constant-acceleration or rotational kinematic relation.

Kinematics - angular velocity displacement

Eq: $\omega^2 = \omega_0^2 + 2 \alpha (\Delta \theta)$
 Law/use: Constant-acceleration or rotational kinematic relation.

Kinematics - tangential acceleration

Eq: $a_t = \alpha r$
 Law/use: Constant-acceleration or rotational kinematic relation.

Kinematics - arc length

Eq: $s = r \theta$
 Law/use: Constant-acceleration or rotational kinematic relation.

Kinematics - centripetal angular relation

Eq: $a_c = \omega^2 r$
 Law/use: Constant-acceleration or rotational kinematic relation.

Electromagnetic and circuit formula appendix**Resistors series**

Eq: $R_{eq} = R_1 + R_2 + \dots$
 Law/use: Combination law for ideal lumped passive components.

Resistors parallel

Eq: $1/R_{eq} = 1/R_1 + 1/R_2 + \dots$
 Law/use: Combination law for ideal lumped passive components.

Capacitors series

Eq: $1/C_{eq} = 1/C_1 + 1/C_2 + \dots$
 Law/use: Combination law for ideal lumped passive components.

Capacitors parallel

Eq: $C_{eq} = C_1 + C_2 + \dots$
 Law/use: Combination law for ideal lumped passive components.

Inductors series uncoupled

Eq: $L_{eq} = L_1 + L_2 + \dots$
 Law/use: Combination law for ideal lumped passive components.

Inductors parallel uncoupled

Eq: $1/L_{eq} = 1/L_1 + 1/L_2 + \dots$
 Law/use: Combination law for ideal lumped passive components.

RC low pass transfer

Eq: $H(j\omega) = 1/(1+j\omega RC)$
 Law/use: Frequency-domain circuit response formula.

RC high pass transfer

Eq: $H(j\omega) = j\omega RC/(1+j\omega RC)$
 Law/use: Frequency-domain circuit response formula.

RL low pass transfer

Eq: $H(j\omega) = R/(R+j\omega L)$
 Law/use: Frequency-domain circuit response formula.

RL high pass transfer

Eq: $H(j\omega) = j\omega L/(R+j\omega L)$
 Law/use: Frequency-domain circuit response formula.

RLC quality factor series

Eq: $Q = \omega_0 L/R$
 Law/use: Frequency-domain circuit response formula.

RLC bandwidth

Eq: $\Delta \omega = R/L$
 Law/use: Frequency-domain circuit response formula.

Half power frequency

Eq: $|H|^2 = \frac{1}{2} |H|_{\max}^2$
 Law/use: Frequency-domain circuit response formula.

Cutoff frequency RC

Eq: $f_c = 1/(2\pi RC)$
 Law/use: Frequency-domain circuit response formula.

Cutoff frequency RL

Eq: $f_c = R/(2\pi L)$
 Law/use: Frequency-domain circuit response formula.

Electric field point charge

Eq: $E = q/(4\pi \epsilon_0 r^2)$
 Law/use: Field of a point charge.

Potential point charge

Eq: $V = q/(4\pi \epsilon_0 r)$
 Law/use: Potential around point charge.

Dipole moment

Eq: $p = q d$
 Law/use: Electric dipole definition.

Dipole far potential

Eq: $V \sim p \cos \theta / (4\pi \epsilon_0 r^2)$
 Law/use: Far-field dipole potential.

Magnetic dipole moment loop

Eq: $m = I A$

Law/use: Magnetic moment of current loop.

Magnetic energy density

Eq: $u_B = B^2 / (2\mu_0)$

Law/use: Energy density in magnetic field.

Electric energy density

Eq: $u_E = 1/2 \epsilon_0 E^2$

Law/use: Energy density in electric field.

Radiation pressure absorbing

Eq: $p_{\text{rad}} = I/c$

Law/use: Pressure on absorbing surface.

Radiation pressure reflecting

Eq: $p_{\text{rad}} = 2I/c$

Law/use: Pressure on reflecting surface.

Wave impedance vacuum

Eq: $Z_0 = \sqrt{\mu_0/\epsilon_0}$

Law/use: Vacuum electromagnetic impedance.

Quantum equation appendix - operators, spectra and information**Orbital angular momentum magnitude**

Eq: $L^2 = l(l+1)\hbar^2$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Orbital angular momentum z

Eq: $L_z = m_l \hbar$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Magnetic moment electron

Eq: $\mu = -g (e/2m) S$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Bohr magneton

Eq: $\mu_B = e \hbar / (2m_e)$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Nuclear magneton

Eq: $\mu_N = e \hbar / (2m_p)$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Fine structure constant

Eq: $\alpha = e^2 / (4\pi \epsilon_0 \hbar c)$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Compton wavelength

Eq: $\lambda_C = h / (m c)$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Reduced Compton wavelength

Eq: $\lambda_{\text{bar}_C} = \hbar / (m c)$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Thermal de Broglie wavelength

Eq: $\lambda_{\text{th}} = h / \sqrt{2\pi m kT}$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Fermi energy free gas

Eq: $E_F = \hbar^2 / (2m) (3\pi^2 n)^{2/3}$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Fermi temperature

Eq: $T_F = E_F / k_B$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Density of states 3D

Eq: $g(E)$ proportional $\sqrt{E}$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Tunneling decay constant

Eq: $\kappa = \sqrt{2m(V-E)} / \hbar$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

WKB quantization

Eq: $\int p \, dq = (n+1/2)\hbar$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Rabi frequency

Eq: $\Omega_R = \text{dipole} \cdot E_0 / \hbar$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Bloch vector length

Eq: $|r| \leq 1$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Von Neumann entropy

Eq: $S = -\text{Tr}(\rho \log \rho)$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Entanglement entropy

Eq: $S_A = -\text{Tr} \rho_A \log \rho_A$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Heisenberg equation

Eq: $dA/dt = (i/\hbar)[H, A] + dA/dt_{\text{explicit}}$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Baker-Campbell-Hausdorff first

Eq: $e^A e^B = \exp(A+B+1/2[A, B]+\dots)$

Law/use: Quantum law, identity, scale or equation used across atomic, condensed matter and quantum information physics.

Relativity and cosmology formula appendix**Lorentz transform x**

Eq: $x' = \gamma(x - vt)$

Law/use: Relativistic or cosmological equation/definition.

Lorentz transform t

Eq: $t' = \gamma(t - vx/c^2)$

Law/use: Relativistic or cosmological equation/definition.

Rapidity

Eq: $\beta = \tanh \phi$

Law/use: Relativistic or cosmological equation/definition.

Proper time

Eq: $d\tau = dt/\gamma$

Law/use: Relativistic or cosmological equation/definition.

Proper acceleration hyperbola

Eq: $x^2 - c^2 t^2 = c^4/a^2$

Law/use: Relativistic or cosmological equation/definition.

Doppler factor

Eq: $D = \sqrt{(1+\beta)/(1-\beta)}$

Law/use: Relativistic or cosmological equation/definition.

Aberration

Eq: $\cos \theta' = (\cos \theta - \beta)/(1 - \beta \cos \theta)$

Law/use: Relativistic or cosmological equation/definition.

Kerr horizon

Eq: $r_{\text{plus}} = GM/c^2 + \sqrt{(GM/c^2)^2 - a^2}$

Law/use: Relativistic or cosmological equation/definition.

Kerr ergosphere

Eq: $g_{tt} = 0$ boundary

Law/use: Relativistic or cosmological equation/definition.

Planck length

Eq: $l_P = \sqrt{\hbar G/c^3}$

Law/use: Relativistic or cosmological equation/definition.

Planck time

Eq: $t_P = \sqrt{\hbar G/c^5}$

Law/use: Relativistic or cosmological equation/definition.

Planck mass

Eq: $m_P = \sqrt{\hbar c/G}$

Law/use: Relativistic or cosmological equation/definition.

Critical density

Eq: $\rho_c = 3H^2/(8\pi G)$

Law/use: Relativistic or cosmological equation/definition.

Density parameter

Eq: $\Omega = \rho/\rho_c$

Law/use: Relativistic or cosmological equation/definition.

Redshift scale factor

Eq: $1+z = a_0/a_{\text{emit}}$

Law/use: Relativistic or cosmological equation/definition.

Luminosity distance

Eq: $F = L/(4\pi d_L^2)$

Law/use: Relativistic or cosmological equation/definition.

Angular diameter distance

Eq: $\theta = \text{size}/d_A$

Law/use: Relativistic or cosmological equation/definition.

Deceleration parameter

Eq: $q = -a \ddot{a}/\dot{a}^2$

Law/use: Relativistic or cosmological equation/definition.

Statistical mechanics, probability and data laws**Mean**

Eq: $\mu = (1/N) \sum x_i$

Law/use: Average value.

Variance population

Eq: $\sigma^2 = (1/N) \sum (x_i - \mu)^2$

Law/use: Spread of values.

Standard deviation

Eq: $\sigma = \sqrt{\text{variance}}$

Law/use: RMS spread.

Covariance

$$\text{Eq: } \text{cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Law/use: Joint variability.

Correlation

$$\text{Eq: } \rho = \text{cov}(X, Y) / (\sigma_X \sigma_Y)$$

Law/use: Normalised covariance.

Normal distribution

$$\text{Eq: } f(x) = 1/(\sigma \sqrt{2\pi}) \exp(-(x - \mu)^2 / (2\sigma^2))$$

Law/use: Gaussian law.

Poisson distribution

$$\text{Eq: } P(k) = \lambda^k \exp(-\lambda) / k!$$

Law/use: Counting distribution.

Binomial distribution

$$\text{Eq: } P(k) = C(n, k) p^k (1-p)^{(n-k)}$$

Law/use: Finite Bernoulli trials.

Exponential distribution

$$\text{Eq: } f(t) = \lambda \exp(-\lambda t)$$

Law/use: Waiting time for Poisson process.

Chi-square

$$\text{Eq: } \chi^2 = \sum((O - E)^2 / E)$$

Law/use: Goodness-of-fit statistic.

Least squares normal equation

$$\text{Eq: } X^T X \beta = X^T y$$

Law/use: Linear regression solution.

Kalman predict

$$\text{Eq: } x_k = F x_{k-1} + B u_k$$

Law/use: State prediction.

Kalman update

$$\text{Eq: } K = P H^T (H P H^T + R)^{-1}$$

Law/use: Optimal linear Gaussian update.

Fokker-Planck equation

$$\text{Eq: } dP/dt = -d(A P)/dx + 1/2 d^2(B P)/dx^2$$

Law/use: Probability density evolution.

Langevin equation

$$\text{Eq: } m \ddot{x} = -\gamma \dot{x} + \eta(t)$$

Law/use: Stochastic dynamics.

Markov property

$$\text{Eq: } P(X_t | \text{history}) = P(X_t | X_{\text{previous}})$$

Law/use: Memoryless state process.

Chapman-Kolmogorov

$$\text{Eq: } P(x, z) = \int P(x, y) P(y, z) dy$$

Law/use: Transition composition law.

Cramer-Rao bound

$$\text{Eq: } \text{var}(\hat{\theta}) \geq 1/I(\theta)$$

Law/use: Estimator variance lower bound.

Fisher information

$$\text{Eq: } I(\theta) = E[(d/d\theta \log L)^2]$$

Law/use: Information in observations.

Maximum entropy

Eq: choose distribution maximizing S under constraints

Law/use: Statistical inference principle.

Constants, units and dimensional reference**Speed of light**

$$\text{Eq: } c = 299792458 \text{ m/s exactly}$$

Law/use: Relativity and EM constant.

Planck constant

$$\text{Eq: } h = 6.62607015 \times 10^{-34} \text{ J}\cdot\text{s exactly}$$

Law/use: Quantum action constant.

Reduced Planck constant

$$\text{Eq: } \hbar = h/(2\pi)$$

Law/use: Quantum angular action.

Elementary charge

$$\text{Eq: } e = 1.602176634 \times 10^{-19} \text{ C exactly}$$

Law/use: Charge quantum.

Boltzmann constant

$$\text{Eq: } k_B = 1.380649 \times 10^{-23} \text{ J/K exactly}$$

Law/use: Thermal energy per kelvin.

Avogadro constant

$$\text{Eq: } N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1} \text{ exactly}$$

Law/use: Entities per mole.

Gas constant

$$\text{Eq: } R = N_A k_B$$

Law/use: Molar gas constant.

Vacuum permittivity

Eq: $\epsilon_0 = 1/(\mu_0 c^2)$

Law/use: Electric constant.

Vacuum permeability

Eq: $\mu_0 \approx 1.25663706212 \times 10^{-6} \text{ N/A}^2$

Law/use: Magnetic constant.

Gravitational constant

Eq: $G \approx 6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$

Law/use: Newtonian gravity constant.

Stefan-Boltzmann constant

Eq: $\sigma = \pi^2 k_B^4 / (60 \hbar^3 c^2)$

Law/use: Blackbody constant.

Fine-structure constant

Eq: $\alpha \approx 1/137.036$

Law/use: EM coupling strength.

Faraday constant

Eq: $F = N_A e$

Law/use: Charge per mole electrons.

Electron rest energy

Eq: $m_e c^2 \approx 0.511 \text{ MeV}$

Law/use: Electron energy scale.

Proton rest energy

Eq: $m_p c^2 \approx 938.27 \text{ MeV}$

Law/use: Proton energy scale.

Standard gravity

Eq: $g_0 = 9.80665 \text{ m/s}^2$

Law/use: Conventional gravity.

Atmosphere

Eq: $1 \text{ atm} = 101325 \text{ Pa}$

Law/use: Standard pressure.

Electron volt

Eq: $1 \text{ eV} = 1.602176634 \times 10^{-19} \text{ J}$

Law/use: Particle/atomic energy unit.

Solar mass

Eq: $M_{\text{sun}} \approx 1.9885 \times 10^30 \text{ kg}$

Law/use: Astronomical mass unit.

Parsec

Eq: $1 \text{ pc} \approx 3.085677581 \times 10^{16} \text{ m}$

Law/use: Astronomical distance unit.

Alphabetic named law checklist - additional physics/engineering laws**Abbe sine condition**

Eq: $n_y \sin u = \text{constant}$

Law/use: Optical imaging condition reducing aberration.

Ampere circuital law

Eq: $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enclosed}}$

Law/use: Magnetic field circulation from current.

Archard wear law

Eq: $V = K W L/H$

Law/use: Wear volume proportional load and sliding distance.

Avrami equation

Eq: $X(t) = 1 - \exp(-k t^n)$

Law/use: Phase transformation kinetics.

Babinet principle

Eq: diffraction pattern of aperture plus complement sums to unobstructed field

Law/use: Diffraction complementarity.

Becquerel decay law

Eq: activity proportional undecayed nuclei

Law/use: Radioactivity law.

Bode sensitivity integral

Eq: integral log sensitivity constrained

Law/use: Feedback control limitation.

Bouguer law

Eq: $I = I_0 \exp(-\alpha x)$

Law/use: Exponential attenuation.

Brillouin function

Eq: magnetization of paramagnet vs field/temp

Law/use: Quantum paramagnetism law.

Cauchy dispersion equation

Eq: $n(\lambda) = A + B/\lambda^2 + C/\lambda^4$

Law/use: Optical dispersion empirical formula.

Curie principle

Eq: effects have at least symmetries of causes

Law/use: Symmetry principle in physics.

Curie-Weiss law

Eq: $\chi = C / (T - T_C)$

Law/use: Magnetic susceptibility near Curie point.

Dulong-Petit law

Eq: $C_m \sim 3R$

Law/use: Solid heat capacity high temperature.

Einstein relation mobility

Eq: $D = \mu kT/q$

Law/use: Diffusion-mobility relation.

Fajans rules

Eq: polarization predicts ionic/covalent character

Law/use: Chemical bonding rule.

Fresnel drag

Eq: $v_{\text{light}} \sim c/n + v(1 - 1/n^2)$

Law/use: Light in moving medium.

Hagen-Poiseuille law

Eq: $Q = \pi r^4 \Delta p / (8 \mu L)$

Law/use: Laminar tube flow.

Hopkinson law

Eq: $\Phi = F_m / R_m$

Law/use: Magnetic circuit analogue of Ohm law.

Joule first law

Eq: $Q = I^2 R t$

Law/use: Resistive heating.

Joule second law

Eq: U of ideal gas depends only on T

Law/use: Ideal gas internal energy property.

Kasha rule

Eq: emission usually from lowest excited state

Law/use: Photochemistry/spectroscopy rule.

Kirchhoff radiation law

Eq: emissivity=absorptivity

Law/use: Thermal radiation equilibrium.

Kramers-Kronig relations

Eq: real and imaginary response parts linked by causality

Law/use: Linear response law.

Lami theorem

Eq: $F_1/\sin a = F_2/\sin b = F_3/\sin c$

Law/use: Equilibrium of three coplanar forces.

Laplace pressure law

Eq: $\Delta p = \gamma(1/R_1 + 1/R_2)$

Law/use: Curved interface pressure.

Lenz law

Eq: induced effects oppose flux change

Law/use: Electromagnetic induction direction.

Lorentz reciprocity

Eq: source-observer interchange symmetry in linear reciprocal media

Law/use: Wave/circuit reciprocity.

Moseley law

Eq: $\sqrt{f} = a(Z - b)$

Law/use: X-ray frequency vs atomic number.

Nernst distribution law

Eq: solute partitions with constant concentration ratio

Law/use: Phase distribution law.

Newton rings

Eq: $r_m^2 = m \lambda R$

Law/use: Thin-film interference rings.

Oersted law

Eq: current produces magnetic field

Law/use: Early EM law.

Paschen-Back effect

Eq: strong-field decoupling of L and S

Law/use: Atomic magnetic splitting regime.

Rayleigh-Jeans law

Eq: $B_\nu = 2\nu^2 kT / c^2$

Law/use: Classical radiation law.

Reciprocal lattice law

Eq: $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi \delta_{ij}$

Law/use: Crystal reciprocal space relation.

Richardson-Dushman equation

Eq: $J = A T^2 \exp(-\phi/kT)$

Law/use: Thermionic emission law.

Sackur-Tetrode equation

Eq: $S = Nk [\ln(V/N(4\pi mU/3Nh^2)^{3/2}) + 5/2]$

Law/use: Ideal monatomic gas entropy.

Sellmeier equation

Eq: $n^2(\lambda) = 1 + \sum B_i \lambda^2 / (\lambda^2 - C_i)$

Law/use: Optical dispersion.

Snell-Descartes law

Eq: $n_1 \sin \theta_1 = n_2 \sin \theta_2$

Law/use: Refraction law.

Steinmetz equation

Eq: $P_h = k f B_{\max}^\alpha$

Law/use: Magnetic core hysteresis loss.

Stokes theorem

Eq: surface integral curl equals boundary circulation

Law/use: Vector calculus law.

Thomson effect

Eq: heat from current in temperature gradient

Law/use: Thermoelectric effect.

Titius-Bode law

Eq: planet spacing empirical relation

Law/use: Historical orbital pattern.

Trouton rule

Eq: $\Delta S_{\text{vap}} \sim 85 \text{ J/mol/K}$

Law/use: Approximate boiling entropy rule.

Wiener-Khinchin theorem

Eq: autocorrelation Fourier transform is power spectral density

Law/use: Random signal theorem.

Young-Dupr equation

Eq: $W_{\text{adh}} = \gamma_{\text{LV}}(1 + \cos \theta)$

Law/use: Wetting adhesion relation.

Young equation wetting

Eq: $\gamma_{\text{SV}} = \gamma_{\text{SL}} + \gamma_{\text{LV}} \cos \theta$

Law/use: Contact angle equilibrium.

Extended equation bank - harmonics, geometry, scaling and dynamics**Standing wave harmonic n=1**

Eq: $f_1 = 1 v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=1

Eq: $E_1 = 1 h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=2

Eq: $f_2 = 2 v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=2

Eq: $E_2 = 4 h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=3

Eq: $f_3 = 3 v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=3

Eq: $E_3 = 9 h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=4

Eq: $f_4 = 4 v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=4

Eq: $E_4 = 16 h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=5

Eq: $f_5 = 5 v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=5

Eq: $E_5 = 25 h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=6

Eq: $f_6 = 6 v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=6

Eq: $E_6 = 36 h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=7

Eq: $f_7 = 7 v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=19

Eq: $E_{19} = 361 \cdot h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=20

Eq: $f_{20} = 20 \cdot v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=20

Eq: $E_{20} = 400 \cdot h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=21

Eq: $f_{21} = 21 \cdot v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=21

Eq: $E_{21} = 441 \cdot h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=22

Eq: $f_{22} = 22 \cdot v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=22

Eq: $E_{22} = 484 \cdot h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=23

Eq: $f_{23} = 23 \cdot v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=23

Eq: $E_{23} = 529 \cdot h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=24

Eq: $f_{24} = 24 \cdot v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=24

Eq: $E_{24} = 576 \cdot h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=25

Eq: $f_{25} = 25 \cdot v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=25

Eq: $E_{25} = 625 \cdot h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=26

Eq: $f_{26} = 26 \cdot v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=26

Eq: $E_{26} = 676 \cdot h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=27

Eq: $f_{27} = 27 \cdot v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=27

Eq: $E_{27} = 729 \cdot h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=28

Eq: $f_{28} = 28 \cdot v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=28

Eq: $E_{28} = 784 \cdot h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=29

Eq: $f_{29} = 29 \cdot v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=29

Eq: $E_{29} = 841 \cdot h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Standing wave harmonic n=30

Eq: $f_{30} = 30 \cdot v / (2L)$

Law/use: Harmonic frequency for string/open pipe with fixed/open boundaries.

Particle in box level n=30

Eq: $E_{30} = 900 \cdot h^2 / (8mL^2)$

Law/use: Quantum energy level in one-dimensional infinite square well.

Circle circumference

Eq: $C = 2\pi \cdot r$

Law/use: Geometry formula used in field, flux, mechanics and thermodynamics calculations.

Circle area

Eq: $A=\pi r^2$

Law/use: Geometry formula used in field, flux, mechanics and thermodynamics calculations.

Sphere surface area

Eq: $A=4\pi r^2$

Law/use: Geometry formula used in field, flux, mechanics and thermodynamics calculations.

Sphere volume

Eq: $V=4/3 \pi r^3$

Law/use: Geometry formula used in field, flux, mechanics and thermodynamics calculations.

Cylinder volume

Eq: $V=\pi r^2 h$

Law/use: Geometry formula used in field, flux, mechanics and thermodynamics calculations.

Cylinder lateral area

Eq: $A=2\pi r h$

Law/use: Geometry formula used in field, flux, mechanics and thermodynamics calculations.

Cone volume

Eq: $V=1/3 \pi r^2 h$

Law/use: Geometry formula used in field, flux, mechanics and thermodynamics calculations.

Torus volume

Eq: $V=2\pi^2 R r^2$

Law/use: Geometry formula used in field, flux, mechanics and thermodynamics calculations.

Gaussian sphere flux area

Eq: $A=4\pi r^2$

Law/use: Geometry formula used in field, flux, mechanics and thermodynamics calculations.

Gaussian cylinder curved area

Eq: $A=2\pi r L$

Law/use: Geometry formula used in field, flux, mechanics and thermodynamics calculations.

Biot number

Eq: $Bi=hL_c/k$

Law/use: Dimensionless group controlling similarity/scaling in transport physics.

Fourier number

Eq: $Fo=\alpha t/L^2$

Law/use: Dimensionless group controlling similarity/scaling in transport physics.

Peclet number

Eq: $Pe=vL/\alpha$

Law/use: Dimensionless group controlling similarity/scaling in transport physics.

Schmidt number

Eq: $Sc=\nu/D$

Law/use: Dimensionless group controlling similarity/scaling in transport physics.

Sherwood number

Eq: $Sh=k_c L/D$

Law/use: Dimensionless group controlling similarity/scaling in transport physics.

Lewis number

Eq: $Le=\alpha/D$

Law/use: Dimensionless group controlling similarity/scaling in transport physics.

Knudsen number

Eq: $Kn=\lambda/L$

Law/use: Dimensionless group controlling similarity/scaling in transport physics.

Eckert number

Eq: $Ec=v^2/(c_p \Delta T)$

Law/use: Dimensionless group controlling similarity/scaling in transport physics.

Cavitation number

Eq: $Ca_v=(p-p_v)/(0.5 \rho v^2)$

Law/use: Dimensionless group controlling similarity/scaling in transport physics.

Euler number

Eq: $Eu=\Delta p/(\rho v^2)$

Law/use: Dimensionless group controlling similarity/scaling in transport physics.

Stanton number

Eq: $St=h/(\rho v c_p)$

Law/use: Dimensionless group controlling similarity/scaling in transport physics.

Magnetic Reynolds number

Eq: $Rm=\mu \sigma v L$

Law/use: Dimensionless group controlling similarity/scaling in transport physics.

Hartmann number

Eq: $Ha=B L \sqrt{\sigma/(\rho \nu)}$

Law/use: Dimensionless group controlling similarity/scaling in transport physics.

Damkohler number

Eq: $Da=\text{reaction rate}/\text{transport rate}$

Law/use: Dimensionless group controlling similarity/scaling in transport physics.

Thiele modulus

Eq: $\phi=L \sqrt{k/D_{eff}}$

Law/use: Dimensionless group controlling similarity/scaling in transport physics.

Ohnesorge number

Eq: $Oh = \mu / \sqrt{\rho \sigma L}$

Law/use: Dimensionless group controlling similarity/scaling in transport physics.

Exponential decay ODE

Eq: $dx/dt = -\lambda x \rightarrow x = x_0 \exp(-\lambda t)$

Law/use: Differential-equation law/formula used across physics.

Logistic law

Eq: $dN/dt = rN(1 - N/K)$

Law/use: Differential-equation law/formula used across physics.

Driven oscillator

Eq: $m \ddot{x} + c \dot{x} + kx = F_0 \cos \omega t$

Law/use: Differential-equation law/formula used across physics.

Damped natural frequency

Eq: $\omega_d = \omega_0 \sqrt{1 - \zeta^2}$

Law/use: Differential-equation law/formula used across physics.

Damping ratio

Eq: $\zeta = c / (2 \sqrt{km})$

Law/use: Differential-equation law/formula used across physics.

Quality factor oscillator

Eq: $Q = 1 / (2\zeta) = m \omega_0 / c$

Law/use: Differential-equation law/formula used across physics.

Diffusion Green scale

Eq: $\text{length} \sim \sqrt{2Dt}$

Law/use: Differential-equation law/formula used across physics.

Wave impedance string

Eq: $Z = T/v = \sqrt{T \mu}$

Law/use: Differential-equation law/formula used across physics.

Characteristic equation oscillator

Eq: $m r^2 + c r + k = 0$

Law/use: Differential-equation law/formula used across physics.

Log decrement

Eq: $\delta = (1/n) \ln(x_0/x_n)$

Law/use: Differential-equation law/formula used across physics.

Thermal and heat-transfer problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Conduction slab [1.1]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [1.2]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [1.3]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [1.4]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [1.5]

$$\text{Eq: } R_{\text{th}} = 1/(hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [1.6]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [1.7]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t/(R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [1.8]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [1.9]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [1.10]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}}/\dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [2.11]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [2.12]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [2.13]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [2.14]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [2.15]

$$\text{Eq: } R_{\text{th}} = 1/(hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [2.16]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [2.17]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t/(R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [2.18]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [2.19]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [2.20]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}}/\dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [3.21]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [3.22]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [3.23]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [3.24]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Electrical power and machines problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

DC power [1.1]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [1.2]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [1.3]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [1.4]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [1.5]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [1.6]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [1.7]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [1.8]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [1.9]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [1.10]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [2.11]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [2.12]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [2.13]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [2.14]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [2.15]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [2.16]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [2.17]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [2.18]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [2.19]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [2.20]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [3.21]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [3.22]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [3.23]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [3.24]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Quantum and atomic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Photon momentum [1.1]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [1.2]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [1.3]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [1.4]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [1.5]

$$\text{Eq: } E_0 = 1/2 \hbar \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [1.6]

$$\text{Eq: } S_i = (\hbar/2) \sigma_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [1.7]

$$\text{Eq: } r_n = a_0 n^2/Z$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [1.8]

$$\text{Eq: } E_n = -13.6 Z^2/n^2 \text{ eV}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [1.9]

$$\text{Eq: } E_J = \hbar^2 J(J+1)/(2I)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [1.10]

$$\text{Eq: } E_v = \hbar \omega (v + 1/2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [2.11]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [2.12]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [2.13]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [2.14]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [2.15]

$$\text{Eq: } E_0 = 1/2 \hbar \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [2.16]

$$\text{Eq: } S_i = (\hbar/2) \sigma_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [2.17]

$$\text{Eq: } r_n = a_0 n^2/Z$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [2.18]

$$\text{Eq: } E_n = -13.6 Z^2/n^2 \text{ eV}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [2.19]

$$\text{Eq: } E_J = \hbar^2 J(J+1)/(2I)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [2.20]

$$\text{Eq: } E_v = \hbar \omega (v + 1/2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [3.21]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [3.22]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [3.23]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [3.24]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fluids and aerodynamic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Lift equation [1.1]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [1.2]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [1.3]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [1.4]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [1.5]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [1.6]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [1.7]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [1.8]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [1.9]

$$\text{Eq: } v = (1/n) R^{2/3} S^{1/2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [1.10]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [2.11]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [2.12]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [2.13]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [2.14]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [2.15]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [2.16]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [2.17]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [2.18]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [2.19]

$$\text{Eq: } v = (1/n) R^{2/3} S^{1/2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [2.20]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [3.21]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [3.22]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [3.23]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [3.24]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Optics and photonics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Numerical aperture [1.1]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [1.2]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [1.3]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [1.4]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam radius [1.5]

Eq: $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian divergence [1.6]

Eq: $\theta = \lambda / (\pi w_0)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cavity free spectral range [1.7]

Eq: $FSR = c / (2L)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fabry-Perot finesse [1.8]

Eq: $F = FSR / \text{linewidth}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon flux [1.9]

Eq: $\Phi = P / (h f)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Irradiance [1.10]

Eq: $E = P/A$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Numerical aperture [2.11]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [2.12]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [2.13]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [2.14]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam radius [2.15]

Eq: $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian divergence [2.16]

Eq: $\theta = \lambda / (\pi w_0)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cavity free spectral range [2.17]

Eq: $FSR = c / (2L)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fabry-Perot finesse [2.18]

Eq: $F = FSR / \text{linewidth}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon flux [2.19]

Eq: $\Phi = P / (h f)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Irradiance [2.20]

Eq: $E = P/A$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Numerical aperture [3.21]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [3.22]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [3.23]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [3.24]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Mathematical physics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Separation of variables [1.1]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [1.2]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [1.3]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [1.4]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spherical Laplacian radial [1.5]

Eq: $(1/r^2) d/dr (r^2 df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cylindrical Laplacian radial [1.6]

Eq: $(1/r) d/dr (r df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Legendre equation [1.7]

Eq: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Bessel equation [1.8]

Eq: $x^2 y'' + x y' + (x^2 - \nu^2)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hermite equation [1.9]

Eq: $y'' - 2xy' + 2ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Laguerre equation [1.10]

Eq: $xy'' + (1-x)y' + ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Separation of variables [2.11]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [2.12]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [2.13]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [2.14]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spherical Laplacian radial [2.15]

Eq: $(1/r^2) d/dr (r^2 df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cylindrical Laplacian radial [2.16]

Eq: $(1/r) d/dr (r df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Legendre equation [2.17]

Eq: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Bessel equation [2.18]

Eq: $x^2 y'' + x y' + (x^2 - \nu^2)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hermite equation [2.19]

Eq: $y'' - 2xy' + 2ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Laguerre equation [2.20]

Eq: $xy'' + (1-x)y' + ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Separation of variables [3.21]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [3.22]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [3.23]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [3.24]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Mechanics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Net force component [1.1]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [1.2]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [1.3]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [1.4]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Static tipping condition [1.5]

$$\text{Eq: line of action of weight crosses pivot edge}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pulley Atwood acceleration [1.6]

$$\text{Eq: } a = (m_2 - m_1)g/(m_1 + m_2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Rocket equation [1.7]

$$\text{Eq: } \Delta v = v_e \ln(m_0/m_f)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thrust [1.8]

$$\text{Eq: } F_T = \dot{m} v_e + (p_e - p_a)A_e$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Specific impulse [1.9]

$$\text{Eq: } I_{sp} = F/(\dot{m} g_0)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gravitational slingshot [1.10]

$$\text{Eq: energy exchange in moving planet frame}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Net force component [2.11]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [2.12]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [2.13]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [2.14]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Static tipping condition [2.15]

$$\text{Eq: line of action of weight crosses pivot edge}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pulley Atwood acceleration [2.16]

$$\text{Eq: } a = (m_2 - m_1)g/(m_1 + m_2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Rocket equation [2.17]

$$\text{Eq: } \Delta v = v_e \ln(m_0/m_f)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thrust [2.18]

$$\text{Eq: } F_T = \dot{m} v_e + (p_e - p_a)A_e$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Specific impulse [2.19]

$$\text{Eq: } I_{sp} = F/(\dot{m} g_0)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gravitational slingshot [2.20]

$$\text{Eq: energy exchange in moving planet frame}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Net force component [3.21]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [3.22]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [3.23]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [3.24]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal and heat-transfer problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Conduction slab [1.1]

$$\text{Eq: } \dot{Q} = kA \Delta T / L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [1.2]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [1.3]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [1.4]

$$\text{Eq: } R_{\text{th}} = L / (kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [1.5]

$$\text{Eq: } R_{\text{th}} = 1 / (hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [1.6]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [1.7]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t / (R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [1.8]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [1.9]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [1.10]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}} / \dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [2.11]

$$\text{Eq: } \dot{Q} = kA \Delta T / L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [2.12]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [2.13]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [2.14]

$$\text{Eq: } R_{\text{th}} = L / (kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [2.15]

$$\text{Eq: } R_{\text{th}} = 1 / (hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [2.16]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [2.17]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t / (R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [2.18]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [2.19]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [2.20]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}} / \dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [3.21]

$$\text{Eq: } \dot{Q} = kA \Delta T / L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [3.22]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [3.23]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [3.24]

$$\text{Eq: } R_{\text{th}} = L / (kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Electrical power and machines problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

DC power [1.1]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [1.2]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [1.3]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [1.4]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [1.5]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [1.6]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [1.7]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [1.8]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [1.9]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [1.10]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [2.11]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [2.12]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [2.13]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [2.14]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [2.15]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [2.16]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [2.17]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [2.18]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [2.19]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [2.20]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [3.21]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [3.22]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [3.23]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [3.24]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Quantum and atomic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Photon momentum [1.1]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [1.2]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [1.3]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [1.4]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [1.5]

$$\text{Eq: } E_0 = 1/2 \hbar \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [1.6]

$$\text{Eq: } S_i = (\hbar/2) \sigma_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [1.7]

$$\text{Eq: } r_n = a_0 n^2/Z$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [1.8]

$$\text{Eq: } E_n = -13.6 Z^2/n^2 \text{ eV}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [1.9]

$$\text{Eq: } E_J = \hbar^2 J(J+1)/(2I)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [1.10]

$$\text{Eq: } E_v = \hbar \omega (v + 1/2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [2.11]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [2.12]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [2.13]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [2.14]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [2.15]

$$\text{Eq: } E_0 = 1/2 \hbar \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [2.16]

$$\text{Eq: } S_i = (\hbar/2) \sigma_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [2.17]

$$\text{Eq: } r_n = a_0 n^2/Z$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [2.18]

$$\text{Eq: } E_n = -13.6 Z^2/n^2 \text{ eV}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [2.19]

$$\text{Eq: } E_J = \hbar^2 J(J+1)/(2I)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [2.20]

$$\text{Eq: } E_v = \hbar \omega (v + 1/2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [3.21]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [3.22]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [3.23]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [3.24]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fluids and aerodynamic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Lift equation [1.1]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [1.2]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [1.3]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [1.4]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [1.5]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [1.6]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [1.7]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [1.8]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [1.9]

$$\text{Eq: } v = (1/n) R^{2/3} S^{1/2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [1.10]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [2.11]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [2.12]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [2.13]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [2.14]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [2.15]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [2.16]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [2.17]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [2.18]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [2.19]

$$\text{Eq: } v = (1/n) R^{2/3} S^{1/2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [2.20]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [3.21]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [3.22]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [3.23]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [3.24]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Optics and photonics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Numerical aperture [1.1]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [1.2]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [1.3]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [1.4]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam radius [1.5]

Eq: $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian divergence [1.6]

Eq: $\theta = \lambda / (\pi w_0)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cavity free spectral range [1.7]

Eq: $FSR = c / (2L)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fabry-Perot finesse [1.8]

Eq: $F = FSR / \text{linewidth}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon flux [1.9]

Eq: $\Phi = P / (h f)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Irradiance [1.10]

Eq: $E = P/A$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Numerical aperture [2.11]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [2.12]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [2.13]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [2.14]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam radius [2.15]

Eq: $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian divergence [2.16]

Eq: $\theta = \lambda / (\pi w_0)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cavity free spectral range [2.17]

Eq: $FSR = c / (2L)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fabry-Perot finesse [2.18]

Eq: $F = FSR / \text{linewidth}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon flux [2.19]

Eq: $\Phi = P / (h f)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Irradiance [2.20]

Eq: $E = P/A$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Numerical aperture [3.21]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [3.22]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [3.23]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [3.24]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Mathematical physics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Separation of variables [1.1]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [1.2]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [1.3]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [1.4]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spherical Laplacian radial [1.5]

Eq: $(1/r^2) d/dr (r^2 df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cylindrical Laplacian radial [1.6]

Eq: $(1/r) d/dr (r df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Legendre equation [1.7]

Eq: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Bessel equation [1.8]

Eq: $x^2 y'' + x y' + (x^2 - \nu^2)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hermite equation [1.9]

Eq: $y'' - 2xy' + 2ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Laguerre equation [1.10]

Eq: $xy'' + (1-x)y' + ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Separation of variables [2.11]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [2.12]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [2.13]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [2.14]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spherical Laplacian radial [2.15]

Eq: $(1/r^2) d/dr (r^2 df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cylindrical Laplacian radial [2.16]

Eq: $(1/r) d/dr (r df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Legendre equation [2.17]

Eq: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Bessel equation [2.18]

Eq: $x^2 y'' + x y' + (x^2 - \nu^2)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hermite equation [2.19]

Eq: $y'' - 2xy' + 2ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Laguerre equation [2.20]

Eq: $xy'' + (1-x)y' + ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Separation of variables [3.21]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [3.22]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [3.23]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [3.24]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Mechanics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Net force component [1.1]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [1.2]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [1.3]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [1.4]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Static tipping condition [1.5]

$$\text{Eq: line of action of weight crosses pivot edge}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pulley Atwood acceleration [1.6]

$$\text{Eq: } a = (m_2 - m_1)g/(m_1 + m_2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Rocket equation [1.7]

$$\text{Eq: } \Delta v = v_e \ln(m_0/m_f)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thrust [1.8]

$$\text{Eq: } F_T = \dot{m} v_e + (p_e - p_a)A_e$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Specific impulse [1.9]

$$\text{Eq: } I_{sp} = F/(\dot{m} g_0)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gravitational slingshot [1.10]

$$\text{Eq: energy exchange in moving planet frame}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Net force component [2.11]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [2.12]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [2.13]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [2.14]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Static tipping condition [2.15]

$$\text{Eq: line of action of weight crosses pivot edge}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pulley Atwood acceleration [2.16]

$$\text{Eq: } a = (m_2 - m_1)g/(m_1 + m_2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Rocket equation [2.17]

$$\text{Eq: } \Delta v = v_e \ln(m_0/m_f)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thrust [2.18]

$$\text{Eq: } F_T = \dot{m} v_e + (p_e - p_a)A_e$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Specific impulse [2.19]

$$\text{Eq: } I_{sp} = F/(\dot{m} g_0)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gravitational slingshot [2.20]

$$\text{Eq: energy exchange in moving planet frame}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Net force component [3.21]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [3.22]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [3.23]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [3.24]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal and heat-transfer problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Conduction slab [1.1]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [1.2]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [1.3]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [1.4]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [1.5]

$$\text{Eq: } R_{\text{th}} = 1/(hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [1.6]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [1.7]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t/(R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [1.8]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [1.9]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [1.10]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}}/\dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [2.11]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [2.12]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [2.13]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [2.14]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [2.15]

$$\text{Eq: } R_{\text{th}} = 1/(hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [2.16]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [2.17]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t/(R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [2.18]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [2.19]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [2.20]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}}/\dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [3.21]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [3.22]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [3.23]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [3.24]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Electrical power and machines problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

DC power [1.1]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [1.2]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [1.3]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [1.4]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [1.5]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [1.6]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [1.7]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [1.8]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [1.9]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [1.10]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [2.11]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [2.12]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [2.13]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [2.14]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [2.15]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [2.16]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [2.17]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [2.18]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [2.19]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [2.20]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [3.21]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [3.22]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [3.23]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [3.24]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Quantum and atomic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Photon momentum [1.1]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [1.2]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [1.3]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [1.4]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [1.5]

$$\text{Eq: } E_0 = 1/2 \hbar \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [1.6]

$$\text{Eq: } S_i = (\hbar/2) \sigma_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [1.7]

$$\text{Eq: } r_n = a_0 n^2/Z$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [1.8]

$$\text{Eq: } E_n = -13.6 Z^2/n^2 \text{ eV}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [1.9]

$$\text{Eq: } E_J = \hbar^2 J(J+1)/(2I)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [1.10]

$$\text{Eq: } E_v = \hbar \omega (v + 1/2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [2.11]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [2.12]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [2.13]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [2.14]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [2.15]

$$\text{Eq: } E_0 = 1/2 \hbar \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [2.16]

$$\text{Eq: } S_i = (\hbar/2) \sigma_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [2.17]

$$\text{Eq: } r_n = a_0 n^2/Z$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [2.18]

$$\text{Eq: } E_n = -13.6 Z^2/n^2 \text{ eV}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [2.19]

$$\text{Eq: } E_J = \hbar^2 J(J+1)/(2I)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [2.20]

$$\text{Eq: } E_v = \hbar \omega (v + 1/2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [3.21]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [3.22]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [3.23]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [3.24]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fluids and aerodynamic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Lift equation [1.1]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [1.2]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [1.3]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [1.4]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [1.5]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [1.6]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [1.7]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [1.8]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [1.9]

$$\text{Eq: } v = (1/n) R^{2/3} S^{1/2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [1.10]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [2.11]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [2.12]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [2.13]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [2.14]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [2.15]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [2.16]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [2.17]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [2.18]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [2.19]

$$\text{Eq: } v = (1/n) R^{2/3} S^{1/2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [2.20]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [3.21]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [3.22]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [3.23]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [3.24]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Optics and photonics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Numerical aperture [1.1]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [1.2]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [1.3]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [1.4]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam radius [1.5]

Eq: $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian divergence [1.6]

Eq: $\theta = \lambda / (\pi w_0)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cavity free spectral range [1.7]

Eq: $FSR = c / (2L)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fabry-Perot finesse [1.8]

Eq: $F = FSR / \text{linewidth}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon flux [1.9]

Eq: $\Phi = P / (h f)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Irradiance [1.10]

Eq: $E = P/A$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Numerical aperture [2.11]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [2.12]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [2.13]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [2.14]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam radius [2.15]

Eq: $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian divergence [2.16]

Eq: $\theta = \lambda / (\pi w_0)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cavity free spectral range [2.17]

Eq: $FSR = c / (2L)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fabry-Perot finesse [2.18]

Eq: $F = FSR / \text{linewidth}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon flux [2.19]

Eq: $\Phi = P / (h f)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Irradiance [2.20]

Eq: $E = P/A$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Numerical aperture [3.21]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [3.22]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [3.23]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [3.24]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Mathematical physics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Separation of variables [1.1]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [1.2]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [1.3]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [1.4]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spherical Laplacian radial [1.5]

Eq: $(1/r^2) d/dr (r^2 df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cylindrical Laplacian radial [1.6]

Eq: $(1/r) d/dr (r df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Legendre equation [1.7]

Eq: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Bessel equation [1.8]

Eq: $x^2 y'' + x y' + (x^2 - \nu^2)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hermite equation [1.9]

Eq: $y'' - 2xy' + 2ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Laguerre equation [1.10]

Eq: $xy'' + (1-x)y' + ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Separation of variables [2.11]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [2.12]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [2.13]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [2.14]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spherical Laplacian radial [2.15]

Eq: $(1/r^2) d/dr (r^2 df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cylindrical Laplacian radial [2.16]

Eq: $(1/r) d/dr (r df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Legendre equation [2.17]

Eq: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Bessel equation [2.18]

Eq: $x^2 y'' + x y' + (x^2 - \nu^2)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hermite equation [2.19]

Eq: $y'' - 2xy' + 2ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Laguerre equation [2.20]

Eq: $xy'' + (1-x)y' + ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Separation of variables [3.21]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [3.22]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [3.23]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [3.24]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Mechanics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Net force component [1.1]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [1.2]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [1.3]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [1.4]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Static tipping condition [1.5]

$$\text{Eq: line of action of weight crosses pivot edge}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pulley Atwood acceleration [1.6]

$$\text{Eq: } a = (m_2 - m_1)g / (m_1 + m_2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Rocket equation [1.7]

$$\text{Eq: } \Delta v = v_e \ln(m_0/m_f)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thrust [1.8]

$$\text{Eq: } F_T = \dot{m} v_e + (p_e - p_a)A_e$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Specific impulse [1.9]

$$\text{Eq: } I_{sp} = F / (\dot{m} g_0)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gravitational slingshot [1.10]

$$\text{Eq: energy exchange in moving planet frame}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Net force component [2.11]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [2.12]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [2.13]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [2.14]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Static tipping condition [2.15]

$$\text{Eq: line of action of weight crosses pivot edge}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pulley Atwood acceleration [2.16]

$$\text{Eq: } a = (m_2 - m_1)g / (m_1 + m_2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Rocket equation [2.17]

$$\text{Eq: } \Delta v = v_e \ln(m_0/m_f)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thrust [2.18]

$$\text{Eq: } F_T = \dot{m} v_e + (p_e - p_a)A_e$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Specific impulse [2.19]

$$\text{Eq: } I_{sp} = F / (\dot{m} g_0)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gravitational slingshot [2.20]

$$\text{Eq: energy exchange in moving planet frame}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Net force component [3.21]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [3.22]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [3.23]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [3.24]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal and heat-transfer problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Conduction slab [1.1]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [1.2]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [1.3]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [1.4]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [1.5]

$$\text{Eq: } R_{\text{th}} = 1/(hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [1.6]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [1.7]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t/(R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [1.8]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [1.9]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [1.10]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}}/\dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [2.11]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [2.12]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [2.13]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [2.14]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [2.15]

$$\text{Eq: } R_{\text{th}} = 1/(hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [2.16]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [2.17]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t/(R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [2.18]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [2.19]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [2.20]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}}/\dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [3.21]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [3.22]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [3.23]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [3.24]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Electrical power and machines problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

DC power [1.1]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [1.2]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [1.3]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [1.4]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [1.5]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [1.6]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [1.7]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [1.8]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [1.9]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [1.10]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [2.11]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [2.12]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [2.13]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [2.14]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [2.15]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [2.16]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [2.17]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [2.18]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [2.19]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [2.20]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [3.21]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [3.22]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [3.23]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [3.24]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Quantum and atomic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Photon momentum [1.1]

Eq: $p = h/\lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [1.2]

Eq: $E[\text{eV}] = 1240/\lambda[\text{nm}]$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [1.3]

Eq: $\Delta p \sim \hbar/(2 \Delta x)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [1.4]

Eq: $T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [1.5]

Eq: $E_0 = 1/2 \hbar \omega$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [1.6]

Eq: $S_i = (\hbar/2)\sigma_i$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [1.7]

Eq: $r_n = a_0 n^2/Z$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [1.8]

Eq: $E_n = -13.6 Z^2/n^2 \text{ eV}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [1.9]

Eq: $E_J = \hbar^2 J(J+1)/(2I)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [1.10]

Eq: $E_v = \hbar \omega (v + 1/2)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [2.11]

Eq: $p = h/\lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [2.12]

Eq: $E[\text{eV}] = 1240/\lambda[\text{nm}]$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [2.13]

Eq: $\Delta p \sim \hbar/(2 \Delta x)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [2.14]

Eq: $T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [2.15]

Eq: $E_0 = 1/2 \hbar \omega$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [2.16]

Eq: $S_i = (\hbar/2)\sigma_i$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [2.17]

Eq: $r_n = a_0 n^2/Z$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [2.18]

Eq: $E_n = -13.6 Z^2/n^2 \text{ eV}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [2.19]

Eq: $E_J = \hbar^2 J(J+1)/(2I)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [2.20]

Eq: $E_v = \hbar \omega (v + 1/2)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [3.21]

Eq: $p = h/\lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [3.22]

Eq: $E[\text{eV}] = 1240/\lambda[\text{nm}]$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [3.23]

Eq: $\Delta p \sim \hbar/(2 \Delta x)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [3.24]

Eq: $T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fluids and aerodynamic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Lift equation [1.1]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [1.2]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [1.3]

$$\text{Eq: } C_p = (p - p_{\infty}) / (\frac{1}{2} \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [1.4]

$$\text{Eq: } C_f = \tau_w / (\frac{1}{2} \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [1.5]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [1.6]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [1.7]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [1.8]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [1.9]

$$\text{Eq: } v = (1/n) R^{2/3} S^{1/2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [1.10]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [2.11]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [2.12]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [2.13]

$$\text{Eq: } C_p = (p - p_{\infty}) / (\frac{1}{2} \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [2.14]

$$\text{Eq: } C_f = \tau_w / (\frac{1}{2} \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [2.15]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [2.16]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [2.17]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [2.18]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [2.19]

$$\text{Eq: } v = (1/n) R^{2/3} S^{1/2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [2.20]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [3.21]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [3.22]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [3.23]

$$\text{Eq: } C_p = (p - p_{\infty}) / (\frac{1}{2} \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [3.24]

$$\text{Eq: } C_f = \tau_w / (\frac{1}{2} \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Optics and photonics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Numerical aperture [1.1]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [1.2]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [1.3]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [1.4]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam radius [1.5]

Eq: $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian divergence [1.6]

Eq: $\theta = \lambda / (\pi w_0)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cavity free spectral range [1.7]

Eq: $FSR = c / (2L)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fabry-Perot finesse [1.8]

Eq: $F = FSR / \text{linewidth}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon flux [1.9]

Eq: $\Phi = P / (h f)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Irradiance [1.10]

Eq: $E = P/A$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Numerical aperture [2.11]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [2.12]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [2.13]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [2.14]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam radius [2.15]

Eq: $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian divergence [2.16]

Eq: $\theta = \lambda / (\pi w_0)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cavity free spectral range [2.17]

Eq: $FSR = c / (2L)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fabry-Perot finesse [2.18]

Eq: $F = FSR / \text{linewidth}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon flux [2.19]

Eq: $\Phi = P / (h f)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Irradiance [2.20]

Eq: $E = P/A$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Numerical aperture [3.21]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [3.22]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [3.23]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [3.24]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Mathematical physics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Separation of variables [1.1]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [1.2]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [1.3]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [1.4]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spherical Laplacian radial [1.5]

Eq: $(1/r^2) d/dr (r^2 df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cylindrical Laplacian radial [1.6]

Eq: $(1/r) d/dr (r df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Legendre equation [1.7]

Eq: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Bessel equation [1.8]

Eq: $x^2 y'' + x y' + (x^2 - \nu^2)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hermite equation [1.9]

Eq: $y'' - 2xy' + 2ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Laguerre equation [1.10]

Eq: $xy'' + (1-x)y' + ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Separation of variables [2.11]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [2.12]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [2.13]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [2.14]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spherical Laplacian radial [2.15]

Eq: $(1/r^2) d/dr (r^2 df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cylindrical Laplacian radial [2.16]

Eq: $(1/r) d/dr (r df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Legendre equation [2.17]

Eq: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Bessel equation [2.18]

Eq: $x^2 y'' + x y' + (x^2 - \nu^2)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hermite equation [2.19]

Eq: $y'' - 2xy' + 2ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Laguerre equation [2.20]

Eq: $xy'' + (1-x)y' + ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Separation of variables [3.21]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [3.22]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [3.23]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [3.24]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Mechanics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Net force component [1.1]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [1.2]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [1.3]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [1.4]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Static tipping condition [1.5]

$$\text{Eq: line of action of weight crosses pivot edge}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pulley Atwood acceleration [1.6]

$$\text{Eq: } a = (m_2 - m_1)g / (m_1 + m_2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Rocket equation [1.7]

$$\text{Eq: } \Delta v = v_e \ln(m_0/m_f)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thrust [1.8]

$$\text{Eq: } F_T = \dot{m} v_e + (p_e - p_a)A_e$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Specific impulse [1.9]

$$\text{Eq: } I_{sp} = F / (\dot{m} g_0)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gravitational slingshot [1.10]

$$\text{Eq: energy exchange in moving planet frame}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Net force component [2.11]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [2.12]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [2.13]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [2.14]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Static tipping condition [2.15]

$$\text{Eq: line of action of weight crosses pivot edge}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pulley Atwood acceleration [2.16]

$$\text{Eq: } a = (m_2 - m_1)g / (m_1 + m_2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Rocket equation [2.17]

$$\text{Eq: } \Delta v = v_e \ln(m_0/m_f)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thrust [2.18]

$$\text{Eq: } F_T = \dot{m} v_e + (p_e - p_a)A_e$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Specific impulse [2.19]

$$\text{Eq: } I_{sp} = F / (\dot{m} g_0)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gravitational slingshot [2.20]

$$\text{Eq: energy exchange in moving planet frame}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Net force component [3.21]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [3.22]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [3.23]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [3.24]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal and heat-transfer problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Conduction slab [1.1]

$$\text{Eq: } \dot{Q} = kA \Delta T / L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [1.2]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [1.3]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [1.4]

$$\text{Eq: } R_{\text{th}} = L / (kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [1.5]

$$\text{Eq: } R_{\text{th}} = 1 / (hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [1.6]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [1.7]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t / (R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [1.8]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [1.9]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [1.10]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}} / \dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [2.11]

$$\text{Eq: } \dot{Q} = kA \Delta T / L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [2.12]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [2.13]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [2.14]

$$\text{Eq: } R_{\text{th}} = L / (kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [2.15]

$$\text{Eq: } R_{\text{th}} = 1 / (hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [2.16]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [2.17]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t / (R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [2.18]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [2.19]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [2.20]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}} / \dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [3.21]

$$\text{Eq: } \dot{Q} = kA \Delta T / L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [3.22]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [3.23]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [3.24]

$$\text{Eq: } R_{\text{th}} = L / (kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Electrical power and machines problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

DC power [1.1]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [1.2]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [1.3]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [1.4]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [1.5]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [1.6]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [1.7]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [1.8]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [1.9]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [1.10]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [2.11]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [2.12]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [2.13]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [2.14]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [2.15]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [2.16]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [2.17]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [2.18]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [2.19]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [2.20]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [3.21]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [3.22]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [3.23]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [3.24]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Quantum and atomic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Photon momentum [1.1]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [1.2]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [1.3]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [1.4]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [1.5]

$$\text{Eq: } E_0 = \frac{1}{2} \hbar \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [1.6]

$$\text{Eq: } S_i = (\hbar/2) \sigma_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [1.7]

$$\text{Eq: } r_n = a_0 n^2/Z$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [1.8]

$$\text{Eq: } E_n = -13.6 Z^2/n^2 \text{ eV}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [1.9]

$$\text{Eq: } E_J = \hbar^2 J(J+1)/(2I)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [1.10]

$$\text{Eq: } E_v = \hbar \omega (v + 1/2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [2.11]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [2.12]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [2.13]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [2.14]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [2.15]

$$\text{Eq: } E_0 = \frac{1}{2} \hbar \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [2.16]

$$\text{Eq: } S_i = (\hbar/2) \sigma_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [2.17]

$$\text{Eq: } r_n = a_0 n^2/Z$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [2.18]

$$\text{Eq: } E_n = -13.6 Z^2/n^2 \text{ eV}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [2.19]

$$\text{Eq: } E_J = \hbar^2 J(J+1)/(2I)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [2.20]

$$\text{Eq: } E_v = \hbar \omega (v + 1/2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [3.21]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [3.22]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [3.23]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [3.24]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fluids and aerodynamic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Lift equation [1.1]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [1.2]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [1.3]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [1.4]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [1.5]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [1.6]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [1.7]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [1.8]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [1.9]

$$\text{Eq: } v = (1/n) R^{2/3} S^{1/2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [1.10]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [2.11]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [2.12]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [2.13]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [2.14]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [2.15]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [2.16]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [2.17]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [2.18]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [2.19]

$$\text{Eq: } v = (1/n) R^{2/3} S^{1/2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [2.20]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [3.21]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [3.22]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [3.23]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [3.24]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Optics and photonics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Numerical aperture [1.1]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [1.2]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [1.3]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [1.4]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam radius [1.5]

Eq: $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian divergence [1.6]

Eq: $\theta = \lambda / (\pi w_0)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cavity free spectral range [1.7]

Eq: $FSR = c / (2L)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fabry-Perot finesse [1.8]

Eq: $F = FSR / \text{linewidth}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon flux [1.9]

Eq: $\Phi = P / (h f)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Irradiance [1.10]

Eq: $E = P/A$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Numerical aperture [2.11]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [2.12]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [2.13]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [2.14]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam radius [2.15]

Eq: $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian divergence [2.16]

Eq: $\theta = \lambda / (\pi w_0)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cavity free spectral range [2.17]

Eq: $FSR = c / (2L)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fabry-Perot finesse [2.18]

Eq: $F = FSR / \text{linewidth}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon flux [2.19]

Eq: $\Phi = P / (h f)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Irradiance [2.20]

Eq: $E = P/A$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Numerical aperture [3.21]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [3.22]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [3.23]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [3.24]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Mathematical physics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Separation of variables [1.1]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [1.2]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [1.3]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [1.4]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spherical Laplacian radial [1.5]

Eq: $(1/r^2) d/dr (r^2 df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cylindrical Laplacian radial [1.6]

Eq: $(1/r) d/dr (r df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Legendre equation [1.7]

Eq: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Bessel equation [1.8]

Eq: $x^2 y'' + x y' + (x^2 - \nu^2)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hermite equation [1.9]

Eq: $y'' - 2xy' + 2ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Laguerre equation [1.10]

Eq: $xy'' + (1-x)y' + ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Separation of variables [2.11]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [2.12]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [2.13]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [2.14]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spherical Laplacian radial [2.15]

Eq: $(1/r^2) d/dr (r^2 df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cylindrical Laplacian radial [2.16]

Eq: $(1/r) d/dr (r df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Legendre equation [2.17]

Eq: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Bessel equation [2.18]

Eq: $x^2 y'' + x y' + (x^2 - \nu^2)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hermite equation [2.19]

Eq: $y'' - 2xy' + 2ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Laguerre equation [2.20]

Eq: $xy'' + (1-x)y' + ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Separation of variables [3.21]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [3.22]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [3.23]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [3.24]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Mechanics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Net force component [1.1]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [1.2]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [1.3]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [1.4]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Static tipping condition [1.5]

$$\text{Eq: line of action of weight crosses pivot edge}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pulley Atwood acceleration [1.6]

$$\text{Eq: } a = (m_2 - m_1)g / (m_1 + m_2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Rocket equation [1.7]

$$\text{Eq: } \Delta v = v_e \ln(m_0/m_f)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thrust [1.8]

$$\text{Eq: } F_T = \dot{m} v_e + (p_e - p_a)A_e$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Specific impulse [1.9]

$$\text{Eq: } I_{sp} = F / (\dot{m} g_0)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gravitational slingshot [1.10]

$$\text{Eq: energy exchange in moving planet frame}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Net force component [2.11]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [2.12]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [2.13]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [2.14]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Static tipping condition [2.15]

$$\text{Eq: line of action of weight crosses pivot edge}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pulley Atwood acceleration [2.16]

$$\text{Eq: } a = (m_2 - m_1)g / (m_1 + m_2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Rocket equation [2.17]

$$\text{Eq: } \Delta v = v_e \ln(m_0/m_f)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thrust [2.18]

$$\text{Eq: } F_T = \dot{m} v_e + (p_e - p_a)A_e$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Specific impulse [2.19]

$$\text{Eq: } I_{sp} = F / (\dot{m} g_0)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gravitational slingshot [2.20]

$$\text{Eq: energy exchange in moving planet frame}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Net force component [3.21]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [3.22]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [3.23]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [3.24]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal and heat-transfer problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Conduction slab [1.1]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [1.2]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [1.3]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [1.4]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [1.5]

$$\text{Eq: } R_{\text{th}} = 1/(hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [1.6]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [1.7]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t/(R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [1.8]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [1.9]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [1.10]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}}/\dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [2.11]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [2.12]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [2.13]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [2.14]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [2.15]

$$\text{Eq: } R_{\text{th}} = 1/(hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [2.16]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [2.17]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t/(R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [2.18]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [2.19]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [2.20]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}}/\dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [3.21]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [3.22]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [3.23]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [3.24]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Electrical power and machines problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

DC power [1.1]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [1.2]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [1.3]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [1.4]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [1.5]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [1.6]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [1.7]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [1.8]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [1.9]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [1.10]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [2.11]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [2.12]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [2.13]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [2.14]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [2.15]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [2.16]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [2.17]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [2.18]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [2.19]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [2.20]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [3.21]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [3.22]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [3.23]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [3.24]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Quantum and atomic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Photon momentum [1.1]

Eq: $p = h/\lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [1.2]

Eq: $E[\text{eV}] = 1240/\lambda[\text{nm}]$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [1.3]

Eq: $\Delta p \sim \hbar/(2 \Delta x)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [1.4]

Eq: $T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [1.5]

Eq: $E_0 = \frac{1}{2} \hbar \omega$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [1.6]

Eq: $S_i = (\hbar/2)\sigma_i$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [1.7]

Eq: $r_n = a_0 n^2/Z$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [1.8]

Eq: $E_n = -13.6 Z^2/n^2 \text{ eV}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [1.9]

Eq: $E_J = \hbar^2 J(J+1)/(2I)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [1.10]

Eq: $E_v = \hbar \omega (v + 1/2)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [2.11]

Eq: $p = h/\lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [2.12]

Eq: $E[\text{eV}] = 1240/\lambda[\text{nm}]$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [2.13]

Eq: $\Delta p \sim \hbar/(2 \Delta x)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [2.14]

Eq: $T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [2.15]

Eq: $E_0 = \frac{1}{2} \hbar \omega$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [2.16]

Eq: $S_i = (\hbar/2)\sigma_i$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [2.17]

Eq: $r_n = a_0 n^2/Z$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [2.18]

Eq: $E_n = -13.6 Z^2/n^2 \text{ eV}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [2.19]

Eq: $E_J = \hbar^2 J(J+1)/(2I)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [2.20]

Eq: $E_v = \hbar \omega (v + 1/2)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [3.21]

Eq: $p = h/\lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [3.22]

Eq: $E[\text{eV}] = 1240/\lambda[\text{nm}]$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [3.23]

Eq: $\Delta p \sim \hbar/(2 \Delta x)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [3.24]

Eq: $T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fluids and aerodynamic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Lift equation [1.1]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [1.2]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [1.3]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [1.4]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [1.5]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [1.6]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [1.7]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [1.8]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [1.9]

$$\text{Eq: } v = (1/n) R^{2/3} S^{1/2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [1.10]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [2.11]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [2.12]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [2.13]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [2.14]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [2.15]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [2.16]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [2.17]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [2.18]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [2.19]

$$\text{Eq: } v = (1/n) R^{2/3} S^{1/2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [2.20]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [3.21]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [3.22]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [3.23]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [3.24]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Optics and photonics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Numerical aperture [1.1]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [1.2]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [1.3]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [1.4]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam radius [1.5]

Eq: $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian divergence [1.6]

Eq: $\theta = \lambda / (\pi w_0)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cavity free spectral range [1.7]

Eq: $FSR = c / (2L)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fabry-Perot finesse [1.8]

Eq: $F = FSR / \text{linewidth}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon flux [1.9]

Eq: $\Phi = P / (h f)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Irradiance [1.10]

Eq: $E = P/A$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Numerical aperture [2.11]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [2.12]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [2.13]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [2.14]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam radius [2.15]

Eq: $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian divergence [2.16]

Eq: $\theta = \lambda / (\pi w_0)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cavity free spectral range [2.17]

Eq: $FSR = c / (2L)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fabry-Perot finesse [2.18]

Eq: $F = FSR / \text{linewidth}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon flux [2.19]

Eq: $\Phi = P / (h f)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Irradiance [2.20]

Eq: $E = P/A$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Numerical aperture [3.21]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [3.22]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [3.23]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [3.24]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Mathematical physics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Separation of variables [1.1]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [1.2]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [1.3]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [1.4]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spherical Laplacian radial [1.5]

Eq: $(1/r^2) d/dr (r^2 df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cylindrical Laplacian radial [1.6]

Eq: $(1/r) d/dr (r df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Legendre equation [1.7]

Eq: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Bessel equation [1.8]

Eq: $x^2 y'' + x y' + (x^2 - \nu^2)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hermite equation [1.9]

Eq: $y'' - 2xy' + 2ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Laguerre equation [1.10]

Eq: $xy'' + (1-x)y' + ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Separation of variables [2.11]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [2.12]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [2.13]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [2.14]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spherical Laplacian radial [2.15]

Eq: $(1/r^2) d/dr (r^2 df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cylindrical Laplacian radial [2.16]

Eq: $(1/r) d/dr (r df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Legendre equation [2.17]

Eq: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Bessel equation [2.18]

Eq: $x^2 y'' + x y' + (x^2 - \nu^2)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hermite equation [2.19]

Eq: $y'' - 2xy' + 2ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Laguerre equation [2.20]

Eq: $xy'' + (1-x)y' + ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Separation of variables [3.21]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [3.22]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [3.23]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [3.24]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Mechanics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Net force component [1.1]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [1.2]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [1.3]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [1.4]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Static tipping condition [1.5]

$$\text{Eq: line of action of weight crosses pivot edge}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pulley Atwood acceleration [1.6]

$$\text{Eq: } a = (m_2 - m_1)g / (m_1 + m_2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Rocket equation [1.7]

$$\text{Eq: } \Delta v = v_e \ln(m_0/m_f)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thrust [1.8]

$$\text{Eq: } F_T = \dot{m} v_e + (p_e - p_a)A_e$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Specific impulse [1.9]

$$\text{Eq: } I_{sp} = F / (\dot{m} g_0)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gravitational slingshot [1.10]

$$\text{Eq: energy exchange in moving planet frame}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Net force component [2.11]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [2.12]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [2.13]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [2.14]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Static tipping condition [2.15]

$$\text{Eq: line of action of weight crosses pivot edge}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pulley Atwood acceleration [2.16]

$$\text{Eq: } a = (m_2 - m_1)g / (m_1 + m_2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Rocket equation [2.17]

$$\text{Eq: } \Delta v = v_e \ln(m_0/m_f)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thrust [2.18]

$$\text{Eq: } F_T = \dot{m} v_e + (p_e - p_a)A_e$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Specific impulse [2.19]

$$\text{Eq: } I_{sp} = F / (\dot{m} g_0)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gravitational slingshot [2.20]

$$\text{Eq: energy exchange in moving planet frame}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Net force component [3.21]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [3.22]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [3.23]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [3.24]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal and heat-transfer problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Conduction slab [1.1]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [1.2]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [1.3]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [1.4]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [1.5]

$$\text{Eq: } R_{\text{th}} = 1/(hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [1.6]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [1.7]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t/(R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [1.8]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [1.9]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [1.10]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}}/\dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [2.11]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [2.12]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [2.13]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [2.14]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [2.15]

$$\text{Eq: } R_{\text{th}} = 1/(hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [2.16]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [2.17]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t/(R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [2.18]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [2.19]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [2.20]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}}/\dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [3.21]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [3.22]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [3.23]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [3.24]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Electrical power and machines problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

DC power [1.1]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [1.2]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [1.3]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [1.4]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [1.5]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [1.6]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [1.7]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [1.8]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [1.9]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [1.10]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [2.11]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [2.12]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [2.13]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [2.14]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [2.15]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [2.16]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [2.17]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [2.18]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [2.19]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [2.20]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [3.21]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [3.22]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [3.23]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [3.24]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Quantum and atomic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Photon momentum [1.1]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [1.2]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [1.3]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [1.4]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [1.5]

$$\text{Eq: } E_0 = 1/2 \hbar \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [1.6]

$$\text{Eq: } S_i = (\hbar/2) \sigma_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [1.7]

$$\text{Eq: } r_n = a_0 n^2/Z$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [1.8]

$$\text{Eq: } E_n = -13.6 Z^2/n^2 \text{ eV}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [1.9]

$$\text{Eq: } E_J = \hbar^2 J(J+1)/(2I)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [1.10]

$$\text{Eq: } E_v = \hbar \omega (v + 1/2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [2.11]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [2.12]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [2.13]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [2.14]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [2.15]

$$\text{Eq: } E_0 = 1/2 \hbar \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [2.16]

$$\text{Eq: } S_i = (\hbar/2) \sigma_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [2.17]

$$\text{Eq: } r_n = a_0 n^2/Z$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [2.18]

$$\text{Eq: } E_n = -13.6 Z^2/n^2 \text{ eV}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [2.19]

$$\text{Eq: } E_J = \hbar^2 J(J+1)/(2I)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [2.20]

$$\text{Eq: } E_v = \hbar \omega (v + 1/2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [3.21]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [3.22]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [3.23]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [3.24]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fluids and aerodynamic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Lift equation [1.1]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [1.2]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [1.3]

$$\text{Eq: } C_p = (p - p_{\infty}) / (\frac{1}{2} \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [1.4]

$$\text{Eq: } C_f = \tau_w / (\frac{1}{2} \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [1.5]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [1.6]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [1.7]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [1.8]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [1.9]

$$\text{Eq: } v = (1/n) R^{2/3} S^{1/2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [1.10]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [2.11]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [2.12]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [2.13]

$$\text{Eq: } C_p = (p - p_{\infty}) / (\frac{1}{2} \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [2.14]

$$\text{Eq: } C_f = \tau_w / (\frac{1}{2} \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [2.15]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [2.16]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [2.17]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [2.18]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [2.19]

$$\text{Eq: } v = (1/n) R^{2/3} S^{1/2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [2.20]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [3.21]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [3.22]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [3.23]

$$\text{Eq: } C_p = (p - p_{\infty}) / (\frac{1}{2} \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [3.24]

$$\text{Eq: } C_f = \tau_w / (\frac{1}{2} \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Optics and photonics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Numerical aperture [1.1]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [1.2]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [1.3]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [1.4]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam radius [1.5]

Eq: $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian divergence [1.6]

Eq: $\theta = \lambda / (\pi w_0)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cavity free spectral range [1.7]

Eq: $FSR = c / (2L)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fabry-Perot finesse [1.8]

Eq: $F = FSR / \text{linewidth}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon flux [1.9]

Eq: $\Phi = P / (h f)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Irradiance [1.10]

Eq: $E = P/A$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Numerical aperture [2.11]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [2.12]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [2.13]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [2.14]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam radius [2.15]

Eq: $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian divergence [2.16]

Eq: $\theta = \lambda / (\pi w_0)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cavity free spectral range [2.17]

Eq: $FSR = c / (2L)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fabry-Perot finesse [2.18]

Eq: $F = FSR / \text{linewidth}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon flux [2.19]

Eq: $\Phi = P / (h f)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Irradiance [2.20]

Eq: $E = P/A$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Numerical aperture [3.21]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [3.22]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [3.23]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [3.24]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Mathematical physics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Separation of variables [1.1]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [1.2]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [1.3]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [1.4]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spherical Laplacian radial [1.5]

Eq: $(1/r^2) d/dr (r^2 df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cylindrical Laplacian radial [1.6]

Eq: $(1/r) d/dr (r df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Legendre equation [1.7]

Eq: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Bessel equation [1.8]

Eq: $x^2 y'' + x y' + (x^2 - \nu^2)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hermite equation [1.9]

Eq: $y'' - 2xy' + 2ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Laguerre equation [1.10]

Eq: $xy'' + (1-x)y' + ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Separation of variables [2.11]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [2.12]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [2.13]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [2.14]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spherical Laplacian radial [2.15]

Eq: $(1/r^2) d/dr (r^2 df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cylindrical Laplacian radial [2.16]

Eq: $(1/r) d/dr (r df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Legendre equation [2.17]

Eq: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Bessel equation [2.18]

Eq: $x^2 y'' + x y' + (x^2 - \nu^2)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hermite equation [2.19]

Eq: $y'' - 2xy' + 2ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Laguerre equation [2.20]

Eq: $xy'' + (1-x)y' + ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Separation of variables [3.21]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [3.22]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [3.23]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [3.24]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Mechanics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Net force component [1.1]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [1.2]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [1.3]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [1.4]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Static tipping condition [1.5]

$$\text{Eq: line of action of weight crosses pivot edge}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pulley Atwood acceleration [1.6]

$$\text{Eq: } a = (m_2 - m_1)g/(m_1 + m_2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Rocket equation [1.7]

$$\text{Eq: } \Delta v = v_e \ln(m_0/m_f)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thrust [1.8]

$$\text{Eq: } F_T = \dot{m} v_e + (p_e - p_a)A_e$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Specific impulse [1.9]

$$\text{Eq: } I_{sp} = F/(\dot{m} g_0)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gravitational slingshot [1.10]

$$\text{Eq: energy exchange in moving planet frame}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Net force component [2.11]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [2.12]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [2.13]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [2.14]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Static tipping condition [2.15]

$$\text{Eq: line of action of weight crosses pivot edge}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pulley Atwood acceleration [2.16]

$$\text{Eq: } a = (m_2 - m_1)g/(m_1 + m_2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Rocket equation [2.17]

$$\text{Eq: } \Delta v = v_e \ln(m_0/m_f)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thrust [2.18]

$$\text{Eq: } F_T = \dot{m} v_e + (p_e - p_a)A_e$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Specific impulse [2.19]

$$\text{Eq: } I_{sp} = F/(\dot{m} g_0)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gravitational slingshot [2.20]

$$\text{Eq: energy exchange in moving planet frame}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Net force component [3.21]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [3.22]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [3.23]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [3.24]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal and heat-transfer problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Conduction slab [1.1]

$$\text{Eq: } \dot{Q} = kA \Delta T / L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [1.2]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [1.3]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [1.4]

$$\text{Eq: } R_{\text{th}} = L / (kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [1.5]

$$\text{Eq: } R_{\text{th}} = 1 / (hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [1.6]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [1.7]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t / (R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [1.8]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [1.9]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [1.10]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}} / \dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [2.11]

$$\text{Eq: } \dot{Q} = kA \Delta T / L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [2.12]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [2.13]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [2.14]

$$\text{Eq: } R_{\text{th}} = L / (kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [2.15]

$$\text{Eq: } R_{\text{th}} = 1 / (hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [2.16]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [2.17]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t / (R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [2.18]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [2.19]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [2.20]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}} / \dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [3.21]

$$\text{Eq: } \dot{Q} = kA \Delta T / L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [3.22]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [3.23]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [3.24]

$$\text{Eq: } R_{\text{th}} = L / (kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Electrical power and machines problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

DC power [1.1]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [1.2]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [1.3]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [1.4]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [1.5]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [1.6]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [1.7]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [1.8]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [1.9]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [1.10]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [2.11]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [2.12]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [2.13]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [2.14]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [2.15]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [2.16]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [2.17]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [2.18]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [2.19]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [2.20]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [3.21]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [3.22]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [3.23]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [3.24]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Quantum and atomic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Photon momentum [1.1]

Eq: $p = h/\lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [1.2]

Eq: $E[\text{eV}] = 1240/\lambda[\text{nm}]$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [1.3]

Eq: $\Delta p \sim \hbar/(2 \Delta x)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [1.4]

Eq: $T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [1.5]

Eq: $E_0 = 1/2 \hbar \omega$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [1.6]

Eq: $S_i = (\hbar/2)\sigma_i$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [1.7]

Eq: $r_n = a_0 n^2/Z$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [1.8]

Eq: $E_n = -13.6 Z^2/n^2 \text{ eV}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [1.9]

Eq: $E_J = \hbar^2 J(J+1)/(2I)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [1.10]

Eq: $E_v = \hbar \omega (v + 1/2)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [2.11]

Eq: $p = h/\lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [2.12]

Eq: $E[\text{eV}] = 1240/\lambda[\text{nm}]$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [2.13]

Eq: $\Delta p \sim \hbar/(2 \Delta x)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [2.14]

Eq: $T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [2.15]

Eq: $E_0 = 1/2 \hbar \omega$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [2.16]

Eq: $S_i = (\hbar/2)\sigma_i$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [2.17]

Eq: $r_n = a_0 n^2/Z$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [2.18]

Eq: $E_n = -13.6 Z^2/n^2 \text{ eV}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [2.19]

Eq: $E_J = \hbar^2 J(J+1)/(2I)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [2.20]

Eq: $E_v = \hbar \omega (v + 1/2)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [3.21]

Eq: $p = h/\lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [3.22]

Eq: $E[\text{eV}] = 1240/\lambda[\text{nm}]$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [3.23]

Eq: $\Delta p \sim \hbar/(2 \Delta x)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [3.24]

Eq: $T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fluids and aerodynamic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Lift equation [1.1]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [1.2]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [1.3]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [1.4]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [1.5]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [1.6]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [1.7]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [1.8]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [1.9]

$$\text{Eq: } v = (1/n) R^{2/3} S^{1/2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [1.10]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [2.11]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [2.12]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [2.13]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [2.14]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [2.15]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [2.16]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [2.17]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [2.18]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [2.19]

$$\text{Eq: } v = (1/n) R^{2/3} S^{1/2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [2.20]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [3.21]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [3.22]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [3.23]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [3.24]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Optics and photonics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Numerical aperture [1.1]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [1.2]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [1.3]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [1.4]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam radius [1.5]

Eq: $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian divergence [1.6]

Eq: $\theta = \lambda / (\pi w_0)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cavity free spectral range [1.7]

Eq: $FSR = c / (2L)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fabry-Perot finesse [1.8]

Eq: $F = FSR / \text{linewidth}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon flux [1.9]

Eq: $\Phi = P / (h f)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Irradiance [1.10]

Eq: $E = P/A$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Numerical aperture [2.11]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [2.12]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [2.13]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [2.14]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam radius [2.15]

Eq: $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian divergence [2.16]

Eq: $\theta = \lambda / (\pi w_0)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cavity free spectral range [2.17]

Eq: $FSR = c / (2L)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fabry-Perot finesse [2.18]

Eq: $F = FSR / \text{linewidth}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon flux [2.19]

Eq: $\Phi = P / (h f)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Irradiance [2.20]

Eq: $E = P/A$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Numerical aperture [3.21]

Eq: $NA = n \sin \theta$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

F-number [3.22]

Eq: $N = f/D$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Depth of focus diffraction [3.23]

Eq: approx $\pm 2 \lambda N^2$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gaussian beam waist [3.24]

Eq: $z_R = \pi w_0^2 / \lambda$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Mathematical physics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Separation of variables [1.1]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [1.2]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [1.3]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [1.4]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spherical Laplacian radial [1.5]

Eq: $(1/r^2) d/dr (r^2 df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cylindrical Laplacian radial [1.6]

Eq: $(1/r) d/dr (r df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Legendre equation [1.7]

Eq: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Bessel equation [1.8]

Eq: $x^2 y'' + x y' + (x^2 - \nu^2)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hermite equation [1.9]

Eq: $y'' - 2xy' + 2ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Laguerre equation [1.10]

Eq: $xy'' + (1-x)y' + ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Separation of variables [2.11]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [2.12]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [2.13]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [2.14]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spherical Laplacian radial [2.15]

Eq: $(1/r^2) d/dr (r^2 df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Cylindrical Laplacian radial [2.16]

Eq: $(1/r) d/dr (r df/dr)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Legendre equation [2.17]

Eq: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Bessel equation [2.18]

Eq: $x^2 y'' + x y' + (x^2 - \nu^2)y = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hermite equation [2.19]

Eq: $y'' - 2xy' + 2ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Laguerre equation [2.20]

Eq: $xy'' + (1-x)y' + ny = 0$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Separation of variables [3.21]

Eq: assume $u(x, t) = X(x)T(t)$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Eigenmode expansion [3.22]

Eq: $u = \sum a_n \phi_n$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Green function solution [3.23]

Eq: $u(x) = \int G(x, x') f(x') dx'$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Residue theorem [3.24]

Eq: $\int_C f dz = 2\pi i \sum \text{residues}$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Mechanics problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Net force component [1.1]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [1.2]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [1.3]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [1.4]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Static tipping condition [1.5]

$$\text{Eq: line of action of weight crosses pivot edge}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pulley Atwood acceleration [1.6]

$$\text{Eq: } a = (m_2 - m_1)g / (m_1 + m_2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Rocket equation [1.7]

$$\text{Eq: } \Delta v = v_e \ln(m_0/m_f)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thrust [1.8]

$$\text{Eq: } F_T = \dot{m} v_e + (p_e - p_a)A_e$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Specific impulse [1.9]

$$\text{Eq: } I_{sp} = F / (\dot{m} g_0)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gravitational slingshot [1.10]

$$\text{Eq: energy exchange in moving planet frame}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Net force component [2.11]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [2.12]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [2.13]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [2.14]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Static tipping condition [2.15]

$$\text{Eq: line of action of weight crosses pivot edge}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pulley Atwood acceleration [2.16]

$$\text{Eq: } a = (m_2 - m_1)g / (m_1 + m_2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Rocket equation [2.17]

$$\text{Eq: } \Delta v = v_e \ln(m_0/m_f)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thrust [2.18]

$$\text{Eq: } F_T = \dot{m} v_e + (p_e - p_a)A_e$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Specific impulse [2.19]

$$\text{Eq: } I_{sp} = F / (\dot{m} g_0)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Gravitational slingshot [2.20]

$$\text{Eq: energy exchange in moving planet frame}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Net force component [3.21]

$$\text{Eq: } \sum F_x = m a_x; \sum F_y = m a_y$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline acceleration frictionless [3.22]

$$\text{Eq: } a = g \sin \theta$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Incline with kinetic friction [3.23]

$$\text{Eq: } a = g(\sin \theta - \mu_k \cos \theta)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Circular bank no friction [3.24]

$$\text{Eq: } \tan \theta = v^2/(rg)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal and heat-transfer problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Conduction slab [1.1]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [1.2]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [1.3]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [1.4]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [1.5]

$$\text{Eq: } R_{\text{th}} = 1/(hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [1.6]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [1.7]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t/(R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [1.8]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [1.9]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [1.10]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}}/\dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [2.11]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [2.12]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [2.13]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [2.14]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance convection [2.15]

$$\text{Eq: } R_{\text{th}} = 1/(hA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Composite resistance [2.16]

$$\text{Eq: } R_{\text{total}} = \sum R_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lumped capacitance [2.17]

$$\text{Eq: } T(t) - T_{\infty} = (T_0 - T_{\infty}) \exp(-t/(R_{\text{th}} C_{\text{th}}))$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Biot lumped criterion [2.18]

$$\text{Eq: } Bi < 0.1 \text{ approximately}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Heat exchanger LMTD [2.19]

$$\text{Eq: } \dot{Q} = U A \Delta T_{\text{lm}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Effectiveness NTU [2.20]

$$\text{Eq: } \epsilon = \dot{Q}_{\text{actual}}/\dot{Q}_{\text{max}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Conduction slab [3.21]

$$\text{Eq: } \dot{Q} = kA \Delta T/L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Convection [3.22]

$$\text{Eq: } \dot{Q} = hA(T_s - T_{\infty})$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Radiation net [3.23]

$$\text{Eq: } \dot{Q} = \epsilon \sigma A(T_s^4 - T_{\text{env}}^4)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Thermal resistance conduction [3.24]

$$\text{Eq: } R_{\text{th}} = L/(kA)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Electrical power and machines problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

DC power [1.1]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [1.2]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [1.3]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [1.4]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [1.5]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [1.6]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [1.7]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [1.8]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [1.9]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [1.10]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [2.11]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [2.12]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [2.13]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [2.14]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Motor torque power [2.15]

$$\text{Eq: } P=T \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Synchronous speed [2.16]

$$\text{Eq: } n_s=120 f/P_{\text{poles}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Slip induction motor [2.17]

$$\text{Eq: } s=(n_s-n_r)/n_s$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Battery energy [2.18]

$$\text{Eq: } E_{\text{Wh}}=V_{\text{nom}} Ah$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

C-rate [2.19]

$$\text{Eq: } I=C_{\text{rate}} \times \text{Capacity}_{\text{Ah}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

State of charge coulomb count [2.20]

$$\text{Eq: } \text{SOC}=\text{SOC}_0 - \int I dt / Q_{\text{nom}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

DC power [3.21]

$$\text{Eq: } P=VI$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Single phase AC [3.22]

$$\text{Eq: } P=V I \cos \phi$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Three phase apparent [3.23]

$$\text{Eq: } S=\sqrt{3}V_{LL} I_L$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Transformer efficiency [3.24]

$$\text{Eq: } \eta=P_{\text{out}}/P_{\text{in}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Quantum and atomic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Photon momentum [1.1]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [1.2]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [1.3]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [1.4]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [1.5]

$$\text{Eq: } E_0 = 1/2 \hbar \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [1.6]

$$\text{Eq: } S_i = (\hbar/2) \sigma_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [1.7]

$$\text{Eq: } r_n = a_0 n^2/Z$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [1.8]

$$\text{Eq: } E_n = -13.6 Z^2/n^2 \text{ eV}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [1.9]

$$\text{Eq: } E_J = \hbar^2 J(J+1)/(2I)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [1.10]

$$\text{Eq: } E_v = \hbar \omega (v + 1/2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [2.11]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [2.12]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [2.13]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [2.14]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Harmonic zero point [2.15]

$$\text{Eq: } E_0 = 1/2 \hbar \omega$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Spin half matrices [2.16]

$$\text{Eq: } S_i = (\hbar/2) \sigma_i$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen radius [2.17]

$$\text{Eq: } r_n = a_0 n^2/Z$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydrogen ion energy [2.18]

$$\text{Eq: } E_n = -13.6 Z^2/n^2 \text{ eV}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Molecular rotor [2.19]

$$\text{Eq: } E_J = \hbar^2 J(J+1)/(2I)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Vibrational oscillator [2.20]

$$\text{Eq: } E_v = \hbar \omega (v + 1/2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon momentum [3.21]

$$\text{Eq: } p = h/\lambda$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Photon energy electronvolt [3.22]

$$\text{Eq: } E[\text{eV}] = 1240/\lambda[\text{nm}]$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Uncertainty estimate [3.23]

$$\text{Eq: } \Delta p \sim \hbar/(2 \Delta x)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Barrier tunnelling approximate [3.24]

$$\text{Eq: } T \sim \exp(-2a \sqrt{2m(V_0 - E)})/\hbar$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Fluids and aerodynamic problem forms - extended equation drill page

Additional named/practical law forms. These pages are intentionally dense so the PDF is a real working compendium rather than a short summary.

Lift equation [1.1]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [1.2]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [1.3]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [1.4]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [1.5]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [1.6]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [1.7]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [1.8]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [1.9]

$$\text{Eq: } v = (1/n) R^{(2/3)} S^{(1/2)}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [1.10]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [2.11]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [2.12]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [2.13]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [2.14]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Boundary layer laminar flat plate [2.15]

$$\text{Eq: } \delta \sim 5x / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Blasius skin friction [2.16]

$$\text{Eq: } C_f, x = 0.664 / \sqrt{\text{Re}_x}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Turbulent flat plate approx [2.17]

$$\text{Eq: } C_f = 0.0592 / \text{Re}_x^{0.2}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Open channel Chezy [2.18]

$$\text{Eq: } v = C \sqrt{R S}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Manning equation [2.19]

$$\text{Eq: } v = (1/n) R^{(2/3)} S^{(1/2)}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Hydraulic diameter [2.20]

$$\text{Eq: } D_h = 4A / P_{\text{wetted}}$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Lift equation [3.21]

$$\text{Eq: } L = \frac{1}{2} \rho v^2 C_L A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Drag equation [3.22]

$$\text{Eq: } D = \frac{1}{2} \rho v^2 C_D A$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Pressure coefficient [3.23]

$$\text{Eq: } C_p = (p - p_{\infty}) / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.

Skin friction coefficient [3.24]

$$\text{Eq: } C_f = \tau_w / (0.5 \rho v^2)$$

Law/use: Use this as a calculation-ready form; check assumptions, boundary conditions, units and sign convention before applying.